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Meron Ka? Evaluating Menstrual Hygiene Practices and Environments Using Exploratory Factor Analysis

In Partial Fulfillment of the Requirements in
Stat 147: Introduction to Multivariate Analysis

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Abstract

Women experience menstruation for a span of over 40 years. Evidently, practicing menstrual hygiene management is essential for women and girls. However, lack of consistent monitoring is still evident due to insufficient basis for addressing the menstrual hygiene needs of menstruators. Hence, this study aims to determine factors affecting the menstrual hygiene management practices of female college undergraduate students in the Greater Manila Area of the Philippines. A total of 679 responses were recorded and used for applying exploratory factor analysis. From the data collected, a total of eight factors were formulated, namely, accessible and functional facility away from home, reliable menstrual material, safe and accessible disposal location, safe and functional home facility, comfort in handling menstrual materials, usability and reusability of menstrual materials, sufficient and accessible supply of menstrual materials, and adequate soap and water. These results signify the needs of menstruating women and girls, which intends to demand for the prioritization of addressing these concerns in order to achieve satisfaction in practicing menstrual hygiene management.

Keywords: Factor Analysis, Menstrual Materials, Menstruation, Menstrual Health and Hygiene (MHH), Menstrual Hygiene Management (MHM), Greater Manila Area (GMA)

Introduction

Background of the Study

According to UNICEF's *Guide to Menstrual Hygiene Materials* (2020), access to safe and dignified menstruation is a basic need for women and girls. Studies also found that there has been a significant increase in awareness towards menstrual health and hygiene (MHH) in the water, sanitation, and hygiene sector (WASH). However, these practices and experiences are still not accompanied by consistent monitoring. This leads to the lack of sufficient evidence to validate a comprehensive set of indicators that could support in monitoring MHH practices.

Furthermore, in a qualitative assessment by Ermory University, UNICEF Philippines, Plan Philippines, and Save the Children Philippines (2020) on the menstruation-related challenges that girls face in school, a number of key recommendations for improving the girls' experiences with menstruation and menstrual hygiene management were enumerated. Some of these are integrating menstruation into reproductive health and hygiene education and providing menstruation education that is age-appropriate, improving WASH facilities and developing systems for maintenance of WASH facilities in public places, establishing systems that enable girls to access absorbent materials comfortably, discreetly, and when in need. To further provide basis on how to improve menstrual hygiene management, this study will explore the factors that would affect the needs for menstrual hygiene management practices of female college undergraduate students.

Objectives of the Study

This study aims to determine factors affecting the menstrual hygiene management practices and environments of female college undergraduate students in the Greater Manila Area in the Philippines. With this, the study aims to meet the following goals and objectives:

- To determine the factors that influence the menstrual hygiene management practices and environments of female college undergraduate students.
- To provide recommendations to the menstruating community on how to improve menstrual hygiene management.
- To give basis for urban planning to help prioritize providing facilities that have safer and more comfortable space for the menstruating community.

Significance of the Study

- The study will be beneficial to the menstruating community as the results can be used as a guide for helping them become aware of what menstrual hygiene practices should be done.
- Parents and family heads may consider the factors to improve their household environment and facilities not just for the comfort of female members but also for the entire household.
- Given the factors, educational agencies can use this as a guide in creating policies and improving facilities for better menstrual-related hygiene and sanitation activities.
- Health practitioners and health agencies could also come up with programs and policies using the factors and results of the study that would help in strengthening the menstrual health awareness of their beneficiaries.
- To optimize the community facilities, especially the water, sanitation, and waste management facilities, community leaders and urban planners can utilize the results of this study in knowing as to which public facilities must be given attention and priority for improvement and better maintenance.

Scope and Limitations

The study aims to determine factors satisfying the menstrual hygiene management practices and environments of female college undergraduate students in the Greater Manila Area (GMA) in the Philippines. To achieve this objective, the researchers conducted an online survey limited only to female undergraduate college students ages 17-25 years old residing in National Capital Region (NCR) and its neighboring provinces - Cavite, Laguna, Rizal, and Bulacan. Hence, the results to be discussed in later sections of this study may only apply to the said respondents/population. The analyses done in this study may not reflect the general opinion and preferences of all the menstruating girls and women in the Philippines.

Review of Related Literature

Menstruation or menses is defined by the UNICEF (2020) Guidance for Monitoring Menstrual Health and Hygiene as "the natural bodily process of releasing blood and associated matter from the uterus through the vagina as part of the menstrual cycle" (p. 8). Given the fact that this is a normal occurrence that happens to some 800 million women and girls everyday, it is rather alarming that menstruation and menstrual hygiene management are still being overlooked around the world (World Bank Group, 2020).

Delving deeper into the concepts of monthly periods, adequate Menstrual Health and Hygiene includes: having a clean menstrual management material that absorbs and collects menstrual blood, being able to change the said material privately and as oftenly possible, having enough water and soap for washing the body, and having safe and convenient facilities for the disposal of used menstrual materials (WHO/UNICEF Joint Monitoring Programme for drinking water, sanitation and hygiene, 2012).

In another publication from the United Nations Children's Fund (UNICEF, 2019) entitled Guide to Menstrual Hygiene Materials, with the goal of providing guidance to their staff members about the most common menstrual hygiene materials, the organization summarized the similarities and differences of the menstrual cloth, reusable pad, disposable pad, menstrual cup, and tampon as seen below in Table 1.

Table 1

Comparison of Common Menstrual Materials

	Menstrual Materials				
Aspects	**Menstrual cloth**	**Reusable pad**	**Disposable pad**	**Menstrual cup/disc**	**Tampon**
Reusability	Maximum of one year	Around one year	Non-reusable	Five to ten years	Non-reusable
Wear time	About two to four hours	About three to six hours	About three to six hours	About six to twelve hours	At most eight hours
Amount per cycle	One	At least five	From 12 to 22	One	From 12 to 22
Level of maintenance	High	High	Low	Medium	Low

Note. Adapted from Guide to Menstrual Hygiene Materials, by UNICEF, 2019, p. 32-33.

Having a deeper understanding about the basic concepts, a number of studies and articles presented below shall help situate and contextualize the problem that menstruating women face today and the factors that influence their menstrual hygiene practices.

In a cross-sectional study by Dasgupta & Sarkar (2008), conducted among 160 girls from West Bengal, India. It was found out that menstrual hygiene standards are severely compromised in locations with not enough basic facilities such as water, bathroom, and privacy. Thus, it is recommended that basic amenities with respect to menstrual hygiene must be provided for them. In addition, girls who were from urban places indicated that their mothers were the main source of information about menstruation, whereas it was their teachers for the rural counterparts.

Similar concerns and challenges regarding menstrual hygiene management in India were also reported in an article by Sinha and Paul (2018). The barriers presented give emphasis on accessibility to menstrual materials and its disposal. Despite the government's initiatives to provide low-cost disposable sanitary menstrual material, a problem still arises in terms of its accessibility and availability. According to National Family Health Survey 4, the most common menstrual material used by Indian women are menstrual cloths that can be reused after washing and drying. Although compostable and non-compostable pads are available in the market, they have limited availability and higher cost that makes it unaffordable for women in low-income households. In addition, sanitary pads remain to be taxed. Disposal mechanisms remain to be a problem in Indian communities especially on the disposal of used napkins as its classification is not yet identified.

A review article of Kaur et.al (2018) also obtained the same results. It was reported that girls in rural areas are not prepared and aware about menstruation resulting in many difficulties and challenges at home, schools, and as well as workplaces. Major findings focus on education, accessibility, water and sanitation facilities, and disposal management. Due to less knowledge on hygiene practices, they are also unaware of infections that may occur during menstruation. High prices of sanitary pads remain a challenge on their accessibility to clean and safe absorbent hence showing their reliance on reusable menstrual cloths. Despite the developments on water and sanitation facilities, the needs of adolescent girls and women were overlooked and ignored. Differences in disposal mechanisms were also observed depending on their location whether at home or outside. When at home, menstrual materials were discarded with domestic wastes while when outside, flushing in toilets is a common practice. With these

results, it is evident that efforts must be made especially by the public authorities to raise awareness on personal and menstrual hygiene practices and to provide better sanitation and disposal facilities.

In addition to the inquiry, Sommer et. al. (2016) recognized the absence of guidance, facilities, and materials for schoolgirls' menstrual management in low and middle-income countries. Evidence points to receiving lack of guidance and education on menstruation, feeling fear, shame, and embarrassment in managing menstruation, and lack of sanitation facilities and spaces to change. With these, they classified five key priorities to guide global, national, and local action towards a knowledgeable and comfortable mindset about menstruation that will enable girls to manage their menses in school in a comfortable, safe, and dignified way. The five key priority includes:

1. Building a strong cross-sectoral evidence base for MHM in schools for prioritization of policies, resource allocation, and programming at scale
2. Developing and disseminating global guidelines for MHM in schools with minimum standards, indicators, and illustrative strategies for adaptation, adoption, and implementation at national and subnational levels
3. Advancing MHM in schools activities through a comprehensive evidence-based advocacy platform that generates policies, funding, and action across sectors and at all levels of government
4. Allocating responsibility to designated government entities for the provision of MHM in schools (including adequate budget and monitoring and evaluation) and the reporting to global channels and constituents
5. Integrating MHM and the capacity and resources to deliver inclusive MHM into the education system.

These key priorities require a diverse range of sectors to ensure optimal involvement and implementation of this advocacy.

In the context of the Philippines, an article celebrating Menstrual Hygiene Day by Save the Children (2020) mentioned that girls and women who menstruate face combined problems in reaching full potential due to lack of access to hygienic and affordable menstrual products, and clean water and toilets. It was also found in a survey conducted by the organization, that the majority of girls are facing difficulty concentrating and participating in their classes when they are in their menstrual period. Thus it was suggested that supporting menstrual health and

hygiene would significantly help millions of girls in the Philippines to get access to education opportunities, protect them from discrimination, and ensure their well-being.

As of now, and after the COVID-19 pandemic, the already inferior treatment of the government towards menstrual health and hygiene became worse. In an online survey from (Plan International, 2020) gathered from 61 WASH and Sexual Reproductive Rights (SRHR) professionals working in 24 different countries (including the Philippines), it was found out that during the COVID-19 pandemic, those who menstruate have experienced restrictions to menstrual products due to shortages or disrupted supply chains, lack of facilities to help change, clean, and dispose of sanitary products, menstrual hygiene management disinformation and misinformation, and reduced access and availability of clean water to help manage periods. With the results, it was recommended that governments invest in inclusive water, sanitation, and hygiene services and facilities to support the people's access to good menstrual hygiene practices. It is also highly suggested that menstrual hygiene management be included in remote learning curriculums to ensure that people can safely manage their menstrual health during the pandemic (Plan International, 2020).

A study by Ellis et.al (2016) that aims to examine Metro Manila, Masbate, and South-Central schools' water, sanitation, and hygiene (WASH) conditions that impact schoolgirl's menstrual hygiene management, found that the main challenges that the students faced are unreliable access to water, lack of disposal mechanisms for menstrual materials, unclean facilities, and insufficient number of latrines. Specifically, they were able to conclude the following results:

- On water accessibility, it was found out that there is inconsistent water availability from faucets for all locations forcing the respondents to resort to secondary water sources which were limited and at times unavailable. These conditions prevent girls from using the school toilets.
- On sanitary conditions, lack of clean and functional toilets is associated with less space for menstrual management. With these, girls controlled their water and food intake at school to avoid using the toilets except for emergencies.
- Regarding privacy conditions, due to fear of being seen by other students, girls do not use toilets with poor structural conditions, specifically having ineffective locks. It was also reported that at times to ensure their privacy, safety and comfort, they go to toilets

with a friend. Girls also go home to change menstrual materials so that their classmates would not know, while some even refrain from attending school due to heavier flows.

- On disposal and waste management, girls tend to dispose of their used sanitary menstrual materials unnoticeably as they keep their menstruation secret from their classmates. Thus, when the trash cans are outside, they dispose of materials when no one is around. Some girls even tried to flush their used menstrual materials resulting in clogged and nonfunctional toilets.

Thus, these show that water, sanitation, and hygiene facilities in schools are still lacking. This can be applied not only to school buildings and facilities but also to all public sanitation facilities. Finally, menstruation is still a social taboo, this explains why in most cases, schoolgirls are ashamed that they are menstruating (Ellis et.al 2016).

Looking closer to the factors that influence the menstrual health practices, In a research article entitled "What are the psychometric properties of a menstrual hygiene management scale: a community-based cross-sectional study", (Ramaiya & Sood, 2020) used exploratory factor analysis and found out that menstrual health management among 450 adolescent girls was influenced by two factors: one about the storage of clean absorbent material and menstrual hygiene, and another about the preparation of clean absorbents, frequency of changing, and disposal.

Meanwhile, a study conducted in Soroti, Uganda by (Henegan et.al, 2020), developed a Menstrual Practice Needs Scale (MPNS-36) to measure the degree to which respondents' menstrual practices and environments meet their needs. Through a systematic review, practices that were identified for the measure were: menstrual materials used, frequency of changing materials, transportation and storage of materials, handwashing during menstrual management, genital and body cleaning, disposal of used materials, and methods of washing and drying materials, including access to a vessel for holding water and the use of soap. Following this, experiences across the range of practices that were included were reflected in the final tool. This study also partially supported the hypothesis that emergent factors would reflect groups of practices and that appraisal of environments would load on separate factors. Furthermore, a four-factor solution for the 28 items was found to be the best fit for the data, supported by two factors for reuse. The subscales for the four factors were 'material and home environment needs', 'transport and school environment needs', 'material reliability concerns', and 'change and disposal insecurity'. Equally important to the four-factors, the subscales for the two factors for reuse were 'reuse needs' and 'reuse insecurity'. What's more to the findings they have

discovered is that greater satisfaction and comfort with menstrual practices do not necessarily equate to fewer worries about their reliability or risks to privacy or safety.

The same scale from the study presented above was used in a confirmatory factor analysis conducted by (Vural & Varışoğlu, 2021) about the Reliability and Validity of the Menstrual Practice Needs Scale (MPNS) among 446 Turkish-speaking girls aged 18-25. There, it was shown that the factors of menstrual practice which were: material and home environment needs, transport and school environment needs, material reliability concerns, and change and disposal insecurity; discovered in a previous exploratory factor analysis were consistent with the results taken from the said location and respondents.

Methodology

The following methods and analyses were employed in order to achieve the objectives of the study.

Method of Data Collection

The researchers collected first-hand data through a survey questionnaire with the aid GoogleForms, an electronic survey platform. The use of online platforms is used due to convenience, practicality, and safety due to the COVID19 pandemic. In gathering respondents, the survey link was posted and distributed through social media platforms such as Facebook, Twitter, and Instagram. To minimize having respondents that are not part of the scope, qualifications were also posted together with the survey link. The duration of the data collection is from January 9 to 10, 2022, with the population of interest being female undergraduate college students whose ages ranged from 17-25 years old, residing in the Greater Manila Area - Metro Manila, Cavite, Laguna, Rizal, and Bulacan, and has experienced menstruation for the last six months.

Variables

The collected data is divided into three sections to distinguish the respondent's qualification, socio-demographic profile, and opinions and preferences on menstrual hygiene management practices. Table 2, Table 3, and Table 4 present the variables for each section.

Table 2

Variables Collected for Qualifications

Variable	Response
Age	Age (in years)
City/Municipality of Residence	City/Municipality
Current School/University	Name of College or University

Table 3

Variables Collected for Socio-Demographic Profile

Variable	Response
Housing Type	● Own House ● Rented House ● Boarding House ● Relative's/Guardian's House ● Others
Household size (all members)	Number of household members
Family monthly income	● Below ₱10,000 ● ₱10,000 to less than ₱20,000 ● ₱20,000 to less than ₱35,000 ● ₱35,000 to less than ₱50,000 ● Above ₱50,000
Water source	● Borewell/ hand pump ● Public tap ● Open well ● Household water supply/piped ● Others
Water availability	● Always available ● Sometimes not available ● Frequently not available
Which of the following toilet essentials do you frequently use?	☐ Dipper (Tabo) ☐ Bidet ☐ Toilet Paper
Menstrual materials used	● Disposable Sanitary Pads ● Menstrual cloth/underwear ● Reusable pad ● Tampons ● Menstrual Disc ● Menstrual Cup
Brand of menstrual materials used	Name of Brand
From whom do you mostly get information about practicing menstrual hygiene management?	● Family ● Friends ● School (Teachers and Educational Materials) ● Multimedia (e.g. TV, Radio, Newspaper, Podcasts) ● Social Media (e.g. Facebook, Twitter, Youtube)

Table 4

Variables Collected for Preferences towards Menstrual Hygiene Management Practices

Menstrual Hygiene Management Practices	
X1	Having the freedom to choose the menstrual materials I want.
X2	The menstrual materials I use are comfortable.
X3	The menstrual materials prevent period stains.
X4	The menstrual materials I use are securely intact while I'm wearing them.
X5	I can easily access new menstrual supplies if I ran out of them.
X6	Having enough supply of my menstrual materials on hand to change them as often as I wanted to.
X7	How long I can possibly wear my menstrual materials.
X8	How long I can use and/or reuse my menstrual materials.
X9	The menstrual materials I use are clean.
X10	The menstrual materials I use are cheap.
X11	The menstrual materials I use block my period odor.
X12	I feel comfortable carrying spare menstrual materials with me outside my home.
X13	I feel comfortable carrying menstrual materials to the place where I change them.
X14	Having a clean place to store my menstrual materials when I am not using them.
X15	My menstrual material storage can't be seen by other people.
X16	I feel comfortable storing my menstrual materials until my next period.
X17	Having enough water to wash my hands and/or vagina as much as I want to.
X18	Having enough soap to wash my hands and/or vagina.
X19	I feel clean during my last period.
X20	Having an accessible disposal location for my used menstrual materials.
X21	I feel comfortable transporting used materials to wash or dispose of them.
X22	Having a disposal location for my used menstrual materials away from animals.
X23	Having a disposal location for my used menstrual materials not seen by other people.
X24	Being able to immediately dispose or clean my menstrual materials.
X25	Disposing my used materials the way I wanted to.
X26	Being able to change my menstrual materials anytime when at home.
X27	Having a clean and satisfying place at home to change my menstrual materials.
X28	Having a private place at home to change my menstrual materials.
X29	Having a private place at home to wash my vagina.

Table 4 (continued)

Variables Collected for Preferences towards Menstrual Hygiene Management Practices

Menstrual Hygiene Management Practices	
X30	Having a place at home to change my menstrual materials safely from physical harm (e.g. animals and unsafe structure).
X31	After I urinate, my menstrual blood wouldn't be seen by any household members.
X32	Being able to change my menstrual materials anytime when away from home
X33	Having a clean and satisfying place in a public facility to change my menstrual materials.
X34	After I urinate, my menstrual blood wouldn't be seen by others (non-household members).
X35	Being able to change my menstrual materials privately in a public facility or any facility away from home.
X36	Having a public facility to change my menstrual material safely from physical harm (e.g. animals and unsafe structure).
X37	Having a public facility where I can wash my vagina.

Data Cleaning

Data cleaning was performed after the data collection. This is done to ensure the reliability and validity of the data before the actual analysis. After data cleaning, the researchers have collected a total of 679 responses. In addition, the indicated location of residence were reclassified according to their respective provinces namely NCR, Bulacan, Cavite, Laguna, and Rizal. The type of menstrual material that the respondents used were also grouped into reusable (comprised of reusable menstrual cloth and pads, menstrual cups, and menstrual discs) and non-reusable (such as disposable sanitary pads and tampons).

Survey Questionnaire

The survey questionnaire is made up of a series of questions and scales that serves as the variables of interest in the study. The construction of this questionnaire was based on the Menstrual Practice Needs Scale (MPNS-36) published in UNICEF's Guidance for Monitoring Menstrual Health and Hygiene. The questionnaire was also adjusted brought on by the current situation of the Philippines in the pandemic. Thus, questions that were created are applicable during the pandemic and were divided into three major sections. Furthermore, the questionnaire formulated in Google Forms was administered through social media platforms such as Facebook, Instagram, and Twitter,

Before the respondent answers the online questionnaire, the respondent is first introduced to the brief background and main objectives of the study. This is followed by a short description of the survey and the data privacy notice for the respondent to ensure their right to confidentiality and to also give assurance that there are no right or wrong answers in the questionnaire since it only assesses opinions and personal preferences. It also includes a consent form section, where any respondent who does not give consent to giving their information will not progress any further into the questionnaire.

The first section of the questionnaire contains the questions that could help the researchers in the data cleaning process after data collection. This is to check whether the respondents met the qualifications to participate in this survey. This section gathers information on the respondent's age, city/municipality of residence, and current college or university. Any record of a respondent who answered the survey form that does not meet these qualifications will be disregarded in the data analysis.

Following this, the second section comprises questions for socio-demographic profiling. This collects the information of the respondent's housing type, household size, family monthly income, water source, and water availability. This section also includes questions regarding the toilet essentials they frequently use, the menstrual material they use, the brand of the menstrual material they use, and the source of information for their menstrual hygiene management practices.

Lastly, the third section, which was divided into three parts in the GoogleForms, is a series of statements about menstrual hygiene management practices where the respondent is asked to indicate how important each statement affects their menstrual hygiene management practices using a scale from 1 (Not important at all) to 10 (Extremely essential).

Accordingly, the three sections contain all the variables of interest in the study with appropriate scales for measurements. These are gathered to capture the necessary and essential responses for the evidence-based analysis of the compiled and translated data.

Analysis of Data

Factor Analysis

To identify the factors affecting the menstrual hygiene management practices exploratory factor analysis was performed. Factor analysis is a multivariate analysis technique

that aims to explain complex and varied relationships that are present among a set of observed variables through a process of uncovering common dimensions or factors that channel the seemingly unrelated variables together.

Before performing the factor analysis, the data was tested for model adequacy using Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy and Bartlett's Test for Sphericity. The results for both tests show that the data is appropriate for factor analysis.

In determining the initial number of factors, visual examination of scree plot, Kaiser's rule and cumulative percentage of total variation criteria were considered. As for the cumulative percentage of total variation criteria, the cut off is set to at least 70%.

In performing factor analysis, the method used in extracting factors is the principal factor solution or the principal axis method. Principal axis method is an exploratory factor analysis extraction method where the measure of the percentage of variance in a dependent variable, that is accounted for by its relationship with a weighted linear combination of a set of independent variables, is used as the initial communality estimate for the variable (American Psychological Association, n.d.). Factor analysis is conducted to questions using Likert Scale, specifically the third section of the survey questionnaire.

Moreover, in finding better results and interpretable factors, factor rotation is used. Orthogonal rotations such as varimax and quartimax rotations and oblique rotations such as oblimin rotation were considered in finding a better set of factors.

Employing these methods and analyses, the researchers started with fourteen factors. Satisfactory results were attained by utilizing the varimax rotation with eight factors.

The above analyses and methods were performed using R software. The following R packages were also used: tidyverse (Wickham et al., 2019) package was used to better facilitate tables and data structures; GPArotation (Bernaards et.al., 2005) package was used in performing factor rotations; psych (Revelle, 2021) package was used in performing the factor analysis; FactorMineR (Lê et. al., 2008) was used in performing exploratory data analysis particularly, principal component analysis; and nFactors (Magis D. & Raiche,G., 2020) package was used for the Non-graphical solutions for Cattell's scree test.

Results and Discussion

Descriptive Statistics

Table 5

Respondents' Profile

Demographic Variables	Response Variable	Frequency	Percentage (%)
Age	17	2	0.3%
	18	47	6.9%
	19	90	13.3%
	20	131	19.3%
	21	200	29.5%
	22	129	19.0%
	23	45	6.6%
	24	17	2.5%
	25	18	2.7%
Residence	NCR	359	52.9%
	Bulacan	71	10.5%
	Cavite	86	12.7%
	Laguna	74	10.9%
	Rizal	89	13.1%
Housing Type	Own House	381	56.1%
	Rented House	116	17.1%
	Boarding House	7	1.0%
	Relative's/Guardian House	166	24.4%
	Others	9	1.3%

Table 5 (continued)

Respondents' Profile

Demographic Variables	Response Variable	Frequency	Percentage (%)
Household Size	1	8	1.2%
	2	47	6.9%
	3	90	13.3%
	4	160	23.6%
	5	158	23.3%
	6	104	15.3%
	7	51	7.5%
	8	26	3.8%
	9	9	1.3%
	10 and above	26	3.8%
Family Monthly Income	below P10,000	60	8.8%
	P10,000 to less than P20,000	147	21.6%
	P20,000 to less than P35,000	146	21.5%
	P35,000 to less than P50,000	105	15.5%
	Above P50,000	221	32.5%
Water Source	Household water supply/piped	582	85.7%
	Open well	2	0.3%
	Bore well	7	1.0%
	Public Tap	62	9.1%
	Others	26	3.8%

Table 5 (continued)

Respondents' Profile

Demographic Variables	Response Variable	Frequency	Percentage (%)
Water Availability	Always Available	632	93.1%
	Sometimes not available	45	6.6%
	Frequently not available	2	0.3%
Menstrual Material Used	Disposable Sanitary Pad/s	360	53.0%
	Menstrual cloth/underwear	5	0.7%
	Menstrual cup	283	41.7%
	Menstrual disc	14	2.1%
	Reusable pad	13	1.9%
	Tampon	4	0.6%
Menstrual Material is disposable/reusable.	Disposable/Non Reusable	364	46.4%
	Reusable	315	53.6%
Frequently used Toilet Essential/s	Dipper (Tabo)	532	78.4%
	Bidet	326	48.0%
	Toilet Paper	218	32.1%
Source of Information about practicing Menstrual Hygiene Management	Family	162	23.9%
	Friends	40	5.9%
	School	43	6.3%
	Multimedia	21	3.1%
	Social Media	413	60.8%

Table 5 shows that the majority of the respondents are highly concentrated between the age range of 20 to 22 years old. Furthermore, most of the respondents reside in NCR with a portion of 52.9%. For the housing type variable, 56.1% of the respondents are living in their own houses. Moreover, the household size of the respondents was mainly concentrated between four (4) to six (6) members. Meanwhile, for the family monthly income, 32.5% of the

respondents receive an income of more than P50,000. A bulk of the respondents also utilizes household water supply/piped at 85.7%. Finally, 93.1% of the respondents answered "always available" in the water availability variable. These are expected as the target population mostly lives in urban areas.

For the variable asking for the menstrual materials that the respondent use, 53.0% of the respondents answered that they use disposable sanitary pads and is followed by 41.7% of the respondents using reusable menstrual cups. After collecting the data on the menstrual materials used by the respondents, the researchers classified the menstrual materials into two groups based on Table 1 in the related literature: disposable/non-reusable and reusable. From the classification, 53.6% of the respondents showed that they are using disposable or non-reusable menstrual materials, while 46.4% showed that they are using reusable menstrual materials. Furthermore, 78.4% of the respondents frequently use dippers (tabo), 48.0% of the respondents also frequently use bidet, and 32.1% of the respondents frequently use toilet paper. Lastly, over 60.8% of the respondents answered that their source of information for practicing menstrual hygiene management is from social media, followed by 23.9% whose source of information is from their family.

Exploratory Factor Analysis

Data Appropriateness/Adequacy

To test if the initial data is appropriate to undergo factor analysis, Bartlett's Test of Sphericity and Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy is conducted.

Table 6

Bartlett's Test of Sphericity

Bartlett's Test of Sphericity	df	Chi-Square	p-value
H0: Variables are orthogonal	666	9147.834	0

Checking the result of the Bartlett's Test of Sphericity, the overall p-value yielded a value of 0, which is less than 0.05. This rejects the null hypothesis that the variables are uncorrelated. That is, there is enough evidence to conclude that the variables are correlated enough. With this, data reduction is applicable for the data particularly, factor analysis.

Table 7

Kaiser-Meyer-Olkin Measure of Sampling Adequacy (KMO-MSA)

Item	MSA	Item	MSA
X1	0.93	X20	0.86
X2	0.91	X21	0.87
X3	0.85	X22	0.84
X4	0.84	X23	0.81
X5	0.78	X24	0.91
X6	0.80	X25	0.91
X7	0.84	X26	0.83
X8	0.80	X27	0.86
X9	0.90	X28	0.76
X10	0.78	X29	0.68
X11	0.91	X30	0.82
X12	0.82	X31	0.68
X13	0.87	X32	0.89
X14	0.89	X33	0.84
X15	0.74	X34	0.69
X16	0.88	X35	0.82
X17	0.75	X36	0.87
X18	0.79	X37	0.85
X19	0.90	**Overall**	0.84

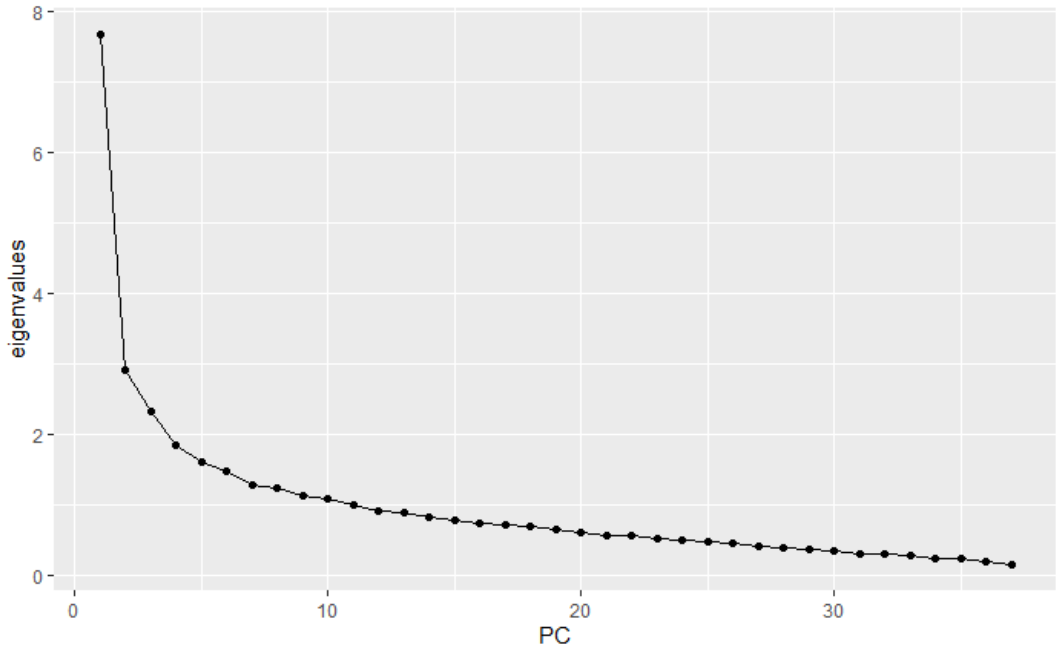
Table 7 above shows the individual KMOs of the 37 variables for the model. The overall KMO of the model is at 0.84, which is considered close to 1 and can be viewed as a "Meritorious" value. Hence, this result also supports the application of factor analysis to the data. However, the researchers have also noted variables X29, X31, and X34 as potential candidates for elimination in the model because of having low KMO measures of 0.68, 0.68, and 0.69 respectively. This is due to the possibility of obtaining correlations between pairs of variables that cannot be explained by the other variables.

Determination of number of factors and variables

Figure 1 presents a scree plot, a graph utilized to select the number of factors that should be considered when conducting the factor analysis (Frey, 2018). Looking closer, the first thing to be noticed is the large scree starting from component 10, this means that the initial number of factors or components is suggested to be 10.

Figure 1

Scree Plot of Eigenvalues



Meanwhile, by conducting the Non-Graphical Cattel's Scree Test as shown in Figure 2, it provides different criteria in choosing the number of factors. Majority of the criteria suggested a 10-factor model. Focusing on the Parallel Analysis or the Kaiser's Rule, the initial number of factors is suggested to be 10 whereas the eigenvalues of the first 10 components is greater than 1.

Figure 2

Non Graphical Solutions to Scree Test

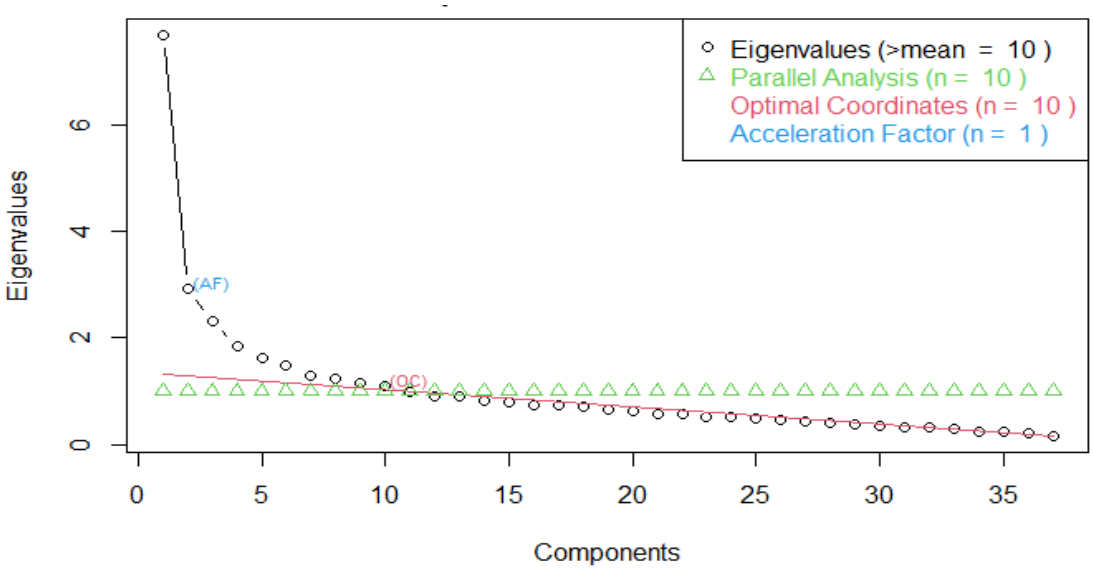


Table 8

Principal Components

Component	Eigenvalue	Percentage of Variance	Cumulative Percentage of Variance
comp 1	7.6798855	20.7564473	20.75645
comp 2	2.9276404	7.9125417	28.66899
comp 3	2.3227403	6.2776765	34.94667
comp 4	1.8480715	4.9947877	39.94145
comp 5	1.6203068	4.3792076	44.32066
comp 6	1.4797163	3.9992332	48.31989
comp 7	1.2972762	3.5061519	51.82605
comp 8	1.2445338	3.3636049	55.18965
comp 9	1.1430279	3.0892646	58.27892
comp 10	1.0936338	2.9557670	61.23468
comp 11	0.9999699	2.7026215	63.93730
comp 12	0.9117704	2.4642443	66.40155
comp 13	0.8933139	2.4143618	68.81591
comp 14	0.8241279	2.2273728	71.04328
comp 15	0.7815845	2.1123906	73.15567
comp 16	0.7473627	2.0198993	75.17557
comp 17	0.7253288	1.9603480	77.13592
comp 18	0.6990832	1.8894139	79.02533
comp 19	0.6492712	1.7547870	80.78012
comp 20	0.6248829	1.6888727	82.46899
comp 21	0.5775655	1.5609878	84.02998
comp 22	0.5671800	1.5329188	85.56290
comp 23	0.5245451	1.4176895	86.98059
comp 24	0.5111534	1.3814957	88.36209
comp 25	0.4839783	1.3080495	89.67014
comp 26	0.4712888	1.2737535	90.94389
comp 27	0.4246186	1.1476179	92.09151
comp 28	0.4046276	1.0935882	93.18510
comp 29	0.3709794	1.0026472	94.18774
comp 30	0.3541994	0.9572957	95.14504

Table 8 (continued)

Principal Components

Component	Eigenvalue	Percentage of Variance	Cumulative Percentage of Variance
comp 31	0.3214511	0.8687867	96.01382
comp 32	0.3162198	0.8546482	96.86847
comp 33	0.2934213	0.7930304	97.66150
comp 34	0.2534349	0.6849593	98.34646
comp 35	0.2374723	0.6418170	98.98828
comp 36	0.2084982	0.5635085	99.55179
comp 37	0.1658384	0.4482119	100

Shown in Table 8 are the eigenvalue of principal components and its corresponding percentage of total variation. Using the cumulative percentage of total variation criterion whereas the cut-off is set to 70%, the researchers decided to initially start with 14 factors given its 71% cumulative percentage of total variation.

In exploring different models, a total of nine runs were performed with a consistent reduction of one (1) factor in each run. The initial run started with 14 factors and 37 variables and the last and ninth run ended with six (6) factors and 29 variables. For the initial run, no rotation, varimax rotation, and quartimax rotation were performed. From the second run until the sixth run, varimax and quartimax rotations were considered. Finally, for the seventh run up to the ninth run, varimax, quartimax, and oblimin rotations were done. In performing the runs, variables X10, X1, X15, X19, and X9 were dropped due to low communality and X31, and X34 were dropped due to low KMO value.

Comparing the results from these runs, the researchers decided to choose the final model generated in the seventh run performed with varimax rotation and the number of factors and variables are eight (8) and 29, respectively. The variables dropped were X1, X9, X10, X15, X19, X31, and X34. The final model was chosen due to its interpretability together with decent communality and KMO values.

Table 9

Bartlett's Test of Sphericity for Final Model

Bartlett's Test of Sphericity	df	Chi-Square	p-value
H0: Variables are orthogonal	435	7586.208	0

Using the final model, the Bartlett's Test of Sphericity as shown in Table 9 remains significant. Thus, rejecting the null hypothesis that the variables are uncorrelated. Given this result, the data is fit for factor analysis.

Table 10

Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy for Final Model

Item	MSA	Item	MSA
X2	0.92	X21	0.88
X3	0.84	X22	0.83
X4	0.84	X23	0.83
X5	0.78	X24	0.90
X6	0.79	X25	0.92
X7	0.83	X26	0.81
X8	0.79	X27	0.84
X9	0.94	X28	0.75
X11	0.90	X29	0.69
X12	0.82	X30	0.79
X13	0.86	X32	0.90
X16	0.87	X33	0.83
X17	0.77	X35	0.81
X18	0.80	X36	0.87
X20	0.85	X37	0.84
Overall		0.84	

As shown in Table 10, the final model has an overall KMO of 0.84 which is interpreted as "Meritorious" such that factor analysis is appropriate for the data. Inspecting the individual KMOs, all of the variables except X29 have KMO values greater than the cut-off of 0.70. The researchers decided to retain this variable due to its interpretability of being included in Factor IV.

Factor Analysis

After the process of elimination of factors and variables and looking at the results of different rotations, the researchers trimmed down the recommended fourteen (14) factors down to eight factors (8) while only 29 out of the 37 questions or variables were retained. This was done by examining whether the factors formed were interpretable while simultaneously consulting the individual variables' communalities and KMO.

As shown in Table 11, the resulting variables are presented together with their corresponding factor loadings for each factor and their communalities. The researchers chose a baseline KMO MSAi value of 0.7 with the exception of variable X29 whose KMO MSAi is 0.691. In addition, the cut-off for the communality was decided to be at 0.3 with variables X24, X16, and X21 considered as exceptions. The said special cases were considered since even if they possess low indices with respect to the said criteria, their inclusion to their corresponding factors were helpful for interpretation.

Table 11

Factor Loadings

Variable	Varimax									KMO MSAi
	I	II	III	IV	V	VI	VII	VIII	Communality	
X35	0.87	0.07	0.20	0.08	0.12	-0.04	0.01	0.03	0.82	0.813
X33	0.77	0.09	0.08	0.03	0.12	0.02	0.10	-0.02	0.63	0.823
X32	0.71	0.10	0.31	0.14	0.18	-0.03	0.15	0.01	0.68	0.897
X37	0.62	0.07	0.10	0.05	-0.03	0.18	0.04	0.16	0.47	0.838
X36	0.61	0.10	0.18	0.13	0.06	0.04	0.03	0.10	0.45	0.867
X4	0.07	0.87	0.06	0.07	0.14	0.15	0.08	0.05	0.81	0.828
X3	0.10	0.80	0.06	0.05	0.08	0.10	0.09	-0.02	0.68	0.838
X2	0.17	0.70	0.04	0.01	0.13	0.12	0.07	0.09	0.56	0.914
X11	0.05	0.47	0.18	0.08	0.03	0.27	0.15	0.01	0.36	0.901
X22	0.14	0.08	0.61	0.03	0.09	-0.03	0.12	0.07	0.42	0.832
X23	0.10	-0.05	0.54	0.07	0.06	0.07	0.02	0.00	0.32	0.828
X25	0.15	0.03	0.50	0.10	0.17	0.05	0.07	0.04	0.33	0.921
X26	0.04	0.10	0.49	0.20	0.18	0.05	-0.04	0.14	0.34	0.809
X24	0.15	0.07	0.47	0.06	0.08	0.05	0.08	0.16	0.29	0.898
X27	0.12	0.13	0.45	0.33	0.13	0.05	-0.03	0.18	0.40	0.842
X20	0.17	0.10	0.43	0.04	0.15	-0.08	0.12	0.20	0.30	0.853

Table 11 (continued)

Factor Loadings

Variable	Varimax									KMO MSAi
	I	II	III	IV	V	VI	VII	VIII	Communality	
X29	0.11	0.03	-0.01	0.86	0.06	0.06	0.05	0.10	0.78	0.691
X28	0.14	0.02	0.18	0.66	0.05	0.05	0.01	0.02	0.49	0.748
X30	0.03	0.09	0.23	0.51	0.01	0.02	-0.01	0.05	0.32	0.789
X12	0.07	0.11	0.18	0.01	0.72	-0.03	0.15	-0.02	0.58	0.812
X13	0.23	0.08	0.19	0.08	0.69	0.04	0.22	-0.02	0.62	0.857
X16	0.01	0.11	0.20	0.12	0.36	0.16	0.00	0.15	0.24	0.862
X21	0.12	0.13	0.23	-0.01	0.33	0.14	0.05	0.03	0.22	0.880
X8	0.05	0.36	0.07	0.04	0.10	0.80	-0.01	0.01	0.78	0.777
X7	0.12	0.38	0.05	0.12	0.13	0.64	0.14	0.08	0.62	0.821
X6	0.07	0.16	0.17	0.04	0.13	0.12	0.77	0.04	0.69	0.787
X5	0.14	0.14	0.07	-0.01	0.20	-0.01	0.63	0.04	0.49	0.774
X17	0.10	0.08	0.15	0.08	0.08	0.05	0.06	0.75	0.62	0.769
X18	0.07	-0.01	0.23	0.08	-0.02	0.01	0.01	0.50	0.32	0.795

Table 12

Percentage of Total Variance Explained

	Factors							
	I	II	III	IV	V	VI	VII	VIII
Proportion Var	0.10	0.09	0.08	0.06	0.05	0.04	0.04	0.03
Cumulative Var	0.10	0.19	0.27	0.32	0.38	0.42	0.46	0.49
Proportion Explained	0.20	0.18	0.16	0.12	0.11	0.09	0.08	0.07
Cumulative Proportion	0.20	0.38	0.54	0.65	0.76	0.85	0.93	1.00

Shown in Table 12 are the percentage of total variance explained of the final eight (8) -factor model. The final model can explain 49% of the total variability. Factor I having the highest percentage of total variance can explain 10% of the total variability. Factor II and Factor III follows with 9% and 8% of the variance explained, respectively. Factor IV and Factor V accounts for 6% and 5% of the variance explained. Both Factor VI and Factor VII can explain 4% of the variance explained. And lastly, Factor 8 can explain 3% of the total variability.

Interpretation of Factors

This study aims to explore and discover what factors do affect the menstrual hygiene management practices and environments of college students from the Greater Manila Area. We shall now look closer and interpret the produced eight (8) factors.

Table 13.1

Factor I - Accessible and Functional Facility away from Home

Variable	Item
X35	Being able to change my menstrual materials privately in a public facility or any facility away from home.
X33	Having a clean and satisfying place in a public facility to change my menstrual materials.
X32	Being able to change my menstrual materials anytime when away from home.
X37	Having a public facility where I can wash my vagina.
X36	Having a public facility to change my menstrual material safely from physical harm (e.g. animals and unsafe structure).

The first factor describes the menstruating community's need for a clean and accessible facility away from home to change their menstrual materials and wash their vagina while being private and safe from physical harm. It could be noted that not being able to find a restroom is a common situation for many, and even if a person finds one, the facility is oftentimes dirty, inconvenient, and unsafe. This can be seen in the cross-sectional study by Dasgupta & Sarkar (2008), where they found out that menstrual hygiene standards are severely compromised in locations with not enough basic facilities such as water, bathroom, and privacy

Table 13.2

Factor II - Reliable Menstrual Material

Variable	Item
X4	The menstrual materials I use are securely intact while I'm wearing them.
X3	The menstrual materials prevent period stains.
X2	The menstrual materials I use are comfortable.
X11	The menstrual materials I use block my period odor.

The second factor deals with the importance of having reliable menstrual material. In other words, either a menstrual cloth or pad, menstrual cup or disc, or tampon, that lets a woman, on her monthly period, continue with her daily activities without worrying about her

menstruation. Sinha and Paul (2018) were able to identify the problems related to this factor. In their study, they were able to find out that sanitary pads remained to be taxed, which makes their prices higher, thus, hindering women in low-income households from availing such products. In addition, menstrual materials also come in various shapes and sizes, thus, making it hard for women to access a product that is suited specifically to their needs.

Table 13.3

Factor III - Safe and Accessible Disposal Location

Variable	Item
X22	Having a disposal location for my used menstrual materials away from animals.
X23	Having a disposal location for my used menstrual materials not seen by other people.
X25	Disposing my used materials the way I wanted to.
X26	Being able to change my menstrual materials anytime when at home.
X24	Being able to immediately dispose or clean my menstrual materials.
X27	Having a clean and satisfying place at home to change my menstrual materials.
X20	Having an accessible disposal location for my used menstrual materials.

The third factor is characterized by having a proper disposal process and location of used menstrual materials. Whether the menstrual material used is reusable or not, disposing of either the contents or the material itself is crucial in assessing menstrual hygiene. Moreover, it is known that the act of menstruation is still considered a taboo in most countries such as the Philippines. This posed a problem to schoolgirls included in a study by Ellis et.al (2016), where they found that due to fear of being seen by other students, girls often dispose of their materials in secret from their classmates. It was also concluded that evident lack of disposal mechanisms for menstrual materials can be found. Finally, disposable menstrual materials are considered as a separate category of waste from the regular biodegradable and non-biodegradable bins since they need additional steps in order to prevent them from exposing people to health risks.

Table 13.4

Factor IV - Safe and Functional Home Facility

Variable	Item
X29	Having a private place at home to wash my vagina.
X28	Having a private place at home to change my menstrual materials.
X30	Having a place at home to change my menstrual materials safely from physical harm (e.g. animals and unsafe structure).

Meanwhile, a home facility that is safe from the invasion of privacy and physical hazard best describes the fourth factor. Even in the context of a person's home, the comfortability of changing menstrual materials may vary depending on the mindset of the household members. In a confirmatory factor analysis conducted by (Vural & Varışoğlu, 2021) about the Reliability and Validity of the Menstrual Practice Needs Scale (MPNS), it was shown that material and home environment needs is one of the factors that mainly affects the menstrual hygiene practices of women.

Table 13.5

Factor V - Comfort in Handling Menstrual Materials

Variable	Item
X12	I feel comfortable carrying spare menstrual materials with me outside my home.
X13	I feel comfortable carrying menstrual materials to the place where I change them.
X16	I feel comfortable storing my menstrual materials until my next period.
X21	I feel comfortable transporting used materials to wash or dispose of them.

Factor five (5) focuses on a person's comfort whenever they handle menstrual materials. This includes carrying, storing, and transporting fresh or used products either for safekeeping or disposal. Due to privacy concerns, women may sometimes feel insecure about carrying menstrual materials whenever they go outside of their homes. In an exploratory factor analysis by Ramaiya & Sood (2020), they discovered that menstrual health management was influenced by two factors – one of which is about the storage of clean absorbent material and menstrual hygiene.

Table 13.6

Factor VI - Usability and Reusability of Menstrual Materials

Variable	Item
X8	How long I can use and/or reuse my menstrual materials.
X7	How long I can possibly wear my menstrual materials.

Subsequently, factor six (6) covers the usability and reusability of the menstrual material. It should be noted that in terms of reusability (setting aside disposable products such as the disposable pad and tampon), menstrual cloths and reusable pads have a lifetime of around one year, while menstrual cups and discs can last up to five (5) to ten (10) years (UNICEF, 2019). Meanwhile, menstrual materials should also be known to have varied wear times, from

only two (2) to four (4) hours for menstrual cloth up to about six (6) to twelve (12) hours for the menstrual cup or disc (UNICEF, 2019). Thus, a person who menstruates values the short-term and long-term longevity of their menstrual material and would need a product that suits their daily routine in the best way in terms of usability and reusability. This was highlighted in a study conducted by (Henegan et.al, 2020), where they stated that factors for reuse namely 'reuse needs' and 'reuse insecurity' were equally important compared to the main factors they found.

Table 13.7

Factor VII - Sufficient and Accessible Supply of Menstrual Materials

Variable	Item
X6	Having enough supply of my menstrual materials on hand to change them as often as I wanted to.
X5	I can easily access new menstrual supplies if I ran out of them.

The seventh factor is denoted by the importance of sufficient and accessible supply. According to the United Nations Children's Fund's (UNICEF, 2019) Guide to Menstrual Hygiene Materials, only one piece is needed per cycle when a menstruating woman is using a menstrual cloth or a menstrual cup or disc. Meanwhile, in terms of disposable products, at least five (5) reusable pads and twelve (12) to twenty-two (22) disposable pads or tampons are utilized by a woman per cycle. Taking these values into account, it can be said that sufficient and accessible supply is essential for a woman to get through a monthly period with ease.

Table 13.8

Factor VIII - Adequate Soap and Water

Variable	Item
X17	Having enough water to wash my hands and/or vagina when I wanted to.
X18	Having enough soap to wash my hands and/or vagina.

Finally, the importance of having enough soap and water is the main idea of the eighth factor. Whether for washing themselves or their reusable menstrual material, soap and water are essential for menstruating women to maintain cleanliness. Note that having enough water and soap for washing the body is one of the main aspects to satisfy in order to achieve an adequate menstrual health and hygiene (WHO/UNICEF Joint Monitoring Programme for drinking water, sanitation, and hygiene, 2012).

Conclusions

Previous studies with the same process and thrusts towards defining the influential factors of menstrual hygiene management highlights a recognizable partition between variables that are associated with menstrual hygiene practices and those related to menstrual hygiene facilities. The researchers entered this undertaking with the same set of initial hypotheses which was then verified through the previously presented statistical analyses. The final factors that were found to influence the menstrual hygiene management practices and environments of female college undergraduate students were: accessible and functional facility away from home, reliable menstrual material, safe and accessible disposal location, safe and functional home facility, comfort in handling menstrual materials, usability and reusability of menstrual materials, sufficient and accessible supply of menstrual materials, and adequate soap and water.

Two of the said factors were about having an accessible and functional location for changing menstrual materials – one inside a person's home and another whenever they are outside. Other factors tackled the importance of having proper disposal and adequate materials for washing. Finally, four factors were concerned about different aspects of menstrual materials such as reliability, usability and reusability, comfort in handling, and sufficient and accessible supply.

Recommendations

With the aforementioned results, the researchers recommend that parents and family heads should take steps in creating a safe space at home for those who are menstruating by dedicating a location for changing and storing menstrual materials and by helping each member be informed about monthly periods. Also, educational agencies are also suggested to create facilities that are accessible and functional for those who are changing their menstrual materials while within their premises, and at the same time formulate programs that would help their audience be aware about the importance of having an ideal menstrual material and facility. Meanwhile, being people with a higher understanding about health risks related to menstrual health and hygiene, health practitioners and agencies are suggested to devise programs and policies that would help ease related concerns such as proper storage and disposal of menstrual materials and choosing the most appropriate product for their needs. Finally, community leaders and urban planners are suggested to prioritize the improvement of public facilities such for menstrual hygiene practices and disposal spaces, and create policies such as lowering the

price of menstrual materials in order to alleviate the burden given by the society to the menstruating community.

For future researchers, it is recommended to add a demographic variable such as religion, which may contribute to the influence of the practices of menstrual hygiene management for women and girls. Confirmatory factor analysis is also suggested as an additional step to further support the factors derived from the exploratory factor analysis. Moreover, the researchers also advise to further add variables for factor analysis since the total variability of the final model is only at 49%. With this, having additional respondents from additional provinces around the country is also recommended to provide wider insight. Future researchers may add questions or statements discussing the internalized stigma in the topic of menstruation to also check whether these may contribute to the evaluation of whether they hinder or promote attaining the essentials of practicing menstrual hygiene management.

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Appendix A

Survey Questionnaire

Meron Ka? Evaluating Menstrual Hygiene Practices using Exploratory Factor Analysis

Good day! We are BS Statistics students from the University of the Philippines Diliman currently enrolled in Statistics 147: Multivariate Analysis. As a partial requirement, we are conducting a study on the factors affecting the menstrual hygiene management practices of female undergraduate college students.

Brief Background of the Study

According to UNICEF, although there has been a significant increase in attention towards menstrual health and hygiene, it still has not been accompanied by consistent monitoring. This study aims to determine factors satisfying the menstrual hygiene management practices and environments of female undergraduate students. With this, this study may help in suggesting safer and more comfortable environments for females to practice better menstrual hygiene management, with ease.

About the Survey

This survey will only take approximately 5 minutes to complete. Note that there are no right or wrong answers to the questions in this survey as they only attempt to assess your opinions and personal preferences.

Data Privacy Notice

By participating in this survey, you consent to process the data you have provided. The researchers are fully committed to protecting the privacy and information of its respondents under the provisions of RA 10173 or the Data Privacy Act of 2012. Any information collected will be strictly kept confidential and will be used solely for the analysis in this study.

By clicking the consent option below, you acknowledge that your participation in this study is voluntary and that you are aware that you may choose to terminate your participation in the study at any time for any reason. If you do not agree with the data privacy statement above you may opt not to answer this survey.

For further queries, kindly contact us:

- Carissa Elaine Varon (cfvaron@up.edu.ph)
- Ray Albert Pujeda (rspujeda@up.edu.ph)
- Ma. Connie Prianes (mvprianes@up.edu.ph)

* Required

1. Email *

2. Age *

Age in years

_____

3. City/Municipality of Residence *

Examples: Antipolo, Rizal; Quezon City

_____

4. Current College/University *

Do not abbreviate. Example: University of the Philippines - Diliman

_____

5. Signed Consent

Mark only one oval.

- ( ) I have read and understood the guidelines and conditions, and I am willing to continue answering this survey. I hereby acknowledge the above statements and give my consent. *Skip to question 6*

Demographic Profile

6. Housing Type *

Mark only one oval.

- ( ) Own House
- ( ) Rented House
- ( ) Boarding House
- ( ) Relative's/Guardian's House
- ( ) Other: _____

7. Household size *

Number of members residing in the household. If living alone, put 1.

_____

8. Family monthly income *

Mark only one oval.

- Below ₱10,000
- ₱10,000 to less than ₱20,000
- ₱20,000 to less than ₱35,000
- ₱35,000 to less than ₱50,000
- Above ₱50,000

9. Water source *

Mark only one oval.

- Bore well/ hand pump
- Public tap
- Open well
- Household water supply/piped
- Others

10. Water availability *

Mark only one oval.

- Always available
- Sometimes not available (not available for at most one whole day per week)
- Frequently not available (not available for at least three whole days per week)

11. Which of the following toilet essentials do you frequently use? *

Check all that apply.

- [ ] Dipper (Tabo)
- [ ] Bidet
- [ ] Toilet Paper

12. Menstrual materials used *

Mark only one oval.

- Disposable Sanitary Pads
- Menstrual Cloth/Underwear
- Reusable Pad
- Tampons
- Menstrual Disc
- Menstrual Cup

13. Brand of menstrual materials used *

Put N/A if unbranded.

_____

14. From whom/where do you mostly receive or get information about practicing menstrual hygiene management? *

Mark only one oval.

- Family
- Friends
- School (Teachers and Educational Materials)
- Multimedia (e.g. TV, Radio, Newspaper, Podcasts)
- Social Media (e.g. Facebook, Twitter, Youtube)

Menstrual Hygiene Management Practices and Environments (Part 1 of 3)

Kindly indicate how important each statement affects your menstrual hygiene management practices using a scale from 1 (Not important at all) to 10 (Extremely essential).

15. Having the freedom to choose the menstrual materials I want. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

16. The menstrual materials I use are comfortable. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

17. The menstrual materials prevent period stains. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

18. The menstrual materials I use are securely intact while I'm wearing them. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

19. I can easily access new menstrual supplies if I ran out of them. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

20. Having enough supply of my menstrual materials on hand to change them as often as I wanted to. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

21. How long I can possibly wear my menstrual materials. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

22. How long I can use and/or reuse my menstrual materials. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

23. The menstrual material I use is clean. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

24. The menstrual materials I use are cheap. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

25. The menstrual materials I use block my period odor. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

Menstrual Hygiene Management Practices and Environments (Part 2 of 3)

(The following statements pertain to different menstrual hygiene practices and environments) Kindly indicate how important each statement affects your menstrual hygiene management practices using a scale from 1 (Not important at all) to 10 (Extremely essential).

26. I feel comfortable carrying spare menstrual materials with me outside my home. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Extremely Essential

27. I feel comfortable carrying menstrual materials to the place where I change them. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Extremely Essential

28. Having a clean place to store my menstrual materials when I am not using them. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Extremely Essential

29. My menstrual material storage can't be seen by other people. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	Extremely Essential

30. I feel comfortable storing my menstrual materials until my next period. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

31. Having enough water to wash my hands and/or vagina when I wanted to. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

32. Having enough soap to wash my hands and/or vagina. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

33. I feel clean during my last period. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

34. Having an accessible disposal location for my used menstrual materials. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

35. I feel comfortable transporting used materials to wash or dispose of them. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

36. Having a disposal location for my used menstrual materials away from animals. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

37. Having a disposal location for my used menstrual materials not seen by other people. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

38. Being able to immediately dispose or clean my menstrual materials. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

39. Disposing my used materials the way I wanted to. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

Menstrual Hygiene Management Practices and Environments (Part 3 of 3)

(The following statements pertain to different menstrual hygiene practices and environments) Kindly indicate how important each statement affects your menstrual hygiene management practices using a scale from 1 (Not important at all) to 10 (Extremely essential).

40. Being able to change my menstrual materials anytime when at home. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

41. Having a clean and satisfying place at home to change my menstrual materials. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

42. Having a private place at home to change my menstrual materials. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

43. Having a private place at home to wash my vagina. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

44. Having a place at home to change my menstrual materials safely from physical harm (e.g. animals and unsafe structure). *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

45. After I urinate, my menstrual blood wouldn't be seen by any household members. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

46. Being able to change my menstrual materials anytime when away from home. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

47. Having a clean and satisfying place in a public facility to change my menstrual materials. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

48. After I urinate, my menstrual blood wouldn't be seen by others (non household members). *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

49. Being able to change my menstrual materials privately in a public facility or any facility away from home. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

50. Having a public facility to change my menstrual material safely from physical harm (e.g. animals and unsafe structure). *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential at all											Extremely Essential

51. Having a public facility where I can wash my vagina. *

Mark only one oval.

	1	2	3	4	5	6	7	8	9	10	
Not Essential At All											Extremely Essential

Raffle Entry

By answering this survey, you are entitled with one raffle entry for a chance to win Php 200.00. To join, kindly indicate your GCash Account Name and GCash Number below. Prizes will be immediately sent to the GCash number of the lucky winners. Please make sure that the GCash Number is valid and matches with the GCash Account Name you provide.

52. GCash Account Name and Number

Example: Mirabel M. - 09123465789

Appendix B

Factor Analysis Models

Initial Run

Table B1.1

14-Factor, 37 Variables, No Rotation

NO ROTATION														
vars	PA1	PA2	PA3	PA4	PA5	PA6	PA7	PA8	PA9	PA10	PA11	PA12	PA13	PA14
x32	0.659	0.271	-0.406	-0.044	-0.004	0.053	-0.072	0.038	-0.013	0.025	-0.071	-0.055	0.053	-0.117
x35	0.6	0.32	-0.542	-0.173	0.127	-0.002	-0.078	-0.058	-0.067	-0.057	0.003	-0.031	0.023	0.055
x13	0.553	0.025	-0.117	0.344	-0.215	0.275	-0.024	-0.237	0.042	0.052	0.043	-0.1	-0.152	-0.041
x7	0.527	-0.469	0.081	-0.154	0.083	0.016	0.053	-0.033	0.325	-0.247	0.085	0.041	-0.14	-0.075
x27	0.521	0.17	0.212	-0.104	-0.208	-0.163	-0.112	-0.063	-0.174	-0.265	0.006	-0.061	0.101	-0.068
x36	0.516	0.226	-0.313	-0.189	0.142	-0.024	-0.016	0.012	-0.035	0.008	0.239	0.129	0.056	0.048
x14	0.515	-0.024	0.015	0.048	-0.078	0.004	0.026	-0.313	0.027	0.1	-0.382	0.239	0.02	-0.157
x2	0.507	-0.48	-0.075	-0.055	0.07	-0.085	-0.025	0.016	-0.162	0.148	-0.091	-0.03	0.024	0.026
x22	0.484	0.244	0.096	0.253	-0.055	-0.193	-0.177	0.312	-0.011	0.06	-0.008	0.298	-0.244	0.124
x26	0.469	0.165	0.256	0.051	-0.186	-0.194	-0.196	-0.098	-0.181	-0.371	0	-0.112	0.133	0.063
x37	0.46	0.14	-0.355	-0.227	0.16	-0.097	0.023	0.023	0.167	0.006	0.117	0.042	0.058	-0.133
x11	0.454	-0.361	0.109	-0.031	0.075	-0.007	-0.033	0.169	0.057	0.041	-0.083	-0.022	0.06	0.039
x24	0.444	0.192	0.103	0.116	0.039	-0.153	0.009	0.076	0.044	-0.017	-0.209	-0.109	-0.034	0.052
x20	0.439	0.178	0.001	0.177	-0.116	-0.178	-0.028	0.121	-0.051	0.103	0.128	0.101	-0.066	-0.18
x6	0.438	-0.15	-0.055	0.302	-0.129	0.159	0.354	0.281	0.047	-0.171	0.002	0.044	0.162	-0.079
x25	0.437	0.194	0.06	0.133	-0.091	-0.069	-0.136	0.114	0.027	-0.075	0.07	-0.124	-0.086	-0.076
x12	0.436	-0.017	-0.017	0.408	-0.182	0.232	-0.046	-0.218	-0.061	0.053	0.134	-0.121	-0.164	0.029
x1	0.407	-0.125	0.057	-0.024	0.042	-0.038	-0.018	-0.151	0.022	-0.075	-0.144	0.114	0.087	0.086
x9	0.405	-0.216	-0.046	-0.024	-0.027	-0.002	0.041	-0.08	0.005	-0.08	-0.015	0.058	-0.123	0.015
x21	0.403	-0.028	0.062	0.217	0.041	0.101	-0.168	-0.065	0.159	0.22	0.266	0.025	0.2	0.081
x19	0.403	-0.126	0.18	0.028	0	0.02	-0.144	-0.065	-0.057	0.126	0.131	0.111	0.168	-0.016
x16	0.401	-0.02	0.225	0.122	-0.012	0.047	-0.003	-0.311	0.076	0.082	-0.003	0.039	0.105	0.097
x28	0.388	0.221	0.224	-0.372	-0.214	0.233	0.023	0.047	-0.005	0.073	-0.029	0.025	-0.015	-0.183
x23	0.381	0.349	0.283	0.168	0.246	0.015	-0.159	0.233	0.191	0.077	-0.172	-0.189	-0.032	-0.007
x30	0.335	0.13	0.264	-0.279	-0.21	0.077	-0.03	0.071	-0.117	-0.013	0.007	0.215	-0.056	0.157
x10	0.228	0.059	0.017	0.051	-0.026	0.137	-0.026	0.165	0.127	0.024	-0.049	-0.064	0.066	0.021
x4	0.556	-0.624	0.034	-0.081	0.073	-0.063	-0.045	0.074	-0.26	0.142	0.002	-0.087	-0.013	-0.033
x3	0.489	-0.544	-0.002	-0.077	0.12	-0.022	-0.042	0.154	-0.271	0.125	0.012	-0.11	-0.035	-0.023
x8	0.43	-0.489	0.117	-0.153	0.129	-0.031	-0.098	-0.042	0.345	-0.215	0.07	0.005	-0.089	0.04
x33	0.524	0.185	-0.566	-0.13	0.124	0.078	-0.01	-0.035	0.032	-0.008	-0.151	-0.074	-0.026	0.178
x15	0.154	0.134	0.334	0.02	0.193	0.126	-0.108	0.062	0.189	0.138	-0.092	-0.005	0.171	0.004
x29	0.363	0.169	0.249	-0.57	-0.397	0.328	0.16	0.085	0.049	0.139	0.028	-0.113	-0.054	0.092
x34	0.341	0.274	0.264	0.018	0.534	0.093	0.29	-0.123	-0.169	-0.013	0.033	0.031	-0.117	-0.046
x31	0.292	0.301	0.409	-0.017	0.489	0.18	0.165	-0.05	-0.176	-0.109	0.096	-0.004	-0.017	0.012
x17	0.373	0.103	0.103	-0.034	-0.128	-0.55	0.445	-0.147	0.13	0.203	0.037	-0.134	-0.04	0.088
x18	0.256	0.162	0.113	-0.026	-0.098	-0.355	0.135	0.002	0.075	0.084	0.091	-0.017	0.077	-0.048
x5	0.38	-0.111	-0.177	0.332	-0.142	0.186	0.394	0.19	-0.054	-0.126	-0.029	0.074	0.121	0.095

Table B1.2

14-Factor, 37 Variables, Varimax Rotation

VARIMAX																
vars	PA3	PA2	PA4	PA5	PA1	PA10	PA9	PA8	PA7	PA6	PA12	PA11	PA13	PA14	communality	KMO MSAi
x35	0.87	0.076	0.055	0.069	0.121	0.155	-0.013	0.037	0.033	0.03	0.059	0.073	-0.005	-0.047	0.824	0.821
x33	0.781	0.116	0.028	-0.01	0.146	-0.006	0.028	0.106	0.093	-0.005	-0.005	0.12	-0.096	-0.202	0.731	0.838
x32	0.714	0.127	0.115	-0.007	0.201	0.152	-0.039	0.189	0.147	0.052	0.086	0.141	0.015	0.128	0.709	0.894
x36	0.648	0.08	0.122	0.153	0.009	0.076	0.064	-0.057	0.049	0.086	0.154	-0.039	0.201	0.017	0.553	0.873
x37	0.63	0.073	0.051	0.041	-0.055	-0.012	0.19	0.046	0.032	0.16	0.028	0.026	0.084	0.163	0.509	0.848
x4	0.06	0.868	0.049	0.017	0.096	0.074	0.161	-0.035	0.074	0.041	0.019	0.056	0.085	0.037	0.82	0.838
x3	0.088	0.802	0.037	0.048	0.067	0.041	0.112	0	0.084	-0.012	0.034	-0.027	0.035	0.035	0.685	0.852
x2	0.157	0.693	-0.001	-0.03	0.07	0.028	0.122	-0.006	0.077	0.07	0.017	0.155	0.073	-0.038	0.568	0.908
x11	0.042	0.494	0.069	-0.006	-0.014	0.048	0.244	0.209	0.153	0.034	0.053	0.078	0.108	-0.03	0.402	0.906
x9	0.146	0.276	0.067	0.03	0.161	0.077	0.26	-0.064	0.103	0.049	0.078	0.144	-0.009	-0.032	0.248	0.904
x29	0.106	0.056	0.907	0.001	0.075	0.039	0.061	0.105	0.034	0.121	-0.098	-0.044	-0.002	-0.067	0.891	0.682
x28	0.142	0.035	0.638	0.102	0.047	0.125	0.031	0.142	0.001	0.033	0.018	0.119	0.031	0.193	0.531	0.762
x30	0.039	0.077	0.514	0.092	-0.039	0.205	0.042	-0.009	0.009	0.02	0.25	0.094	0.091	-0.125	0.421	0.815
x34	0.137	0.044	0.032	0.798	0.058	0.008	0.013	0.12	0.027	0.11	0.044	0.096	0.006	0.018	0.7	0.693
x31	0.038	-0.004	0.117	0.783	0.003	0.145	0.019	0.189	0.018	-0.012	0.02	-0.014	0.1	-0.014	0.697	0.676
x13	0.219	0.094	0.075	-0.003	0.694	0.088	0.081	0.089	0.203	0.007	0.061	0.177	0.117	0.037	0.657	0.867
x12	0.071	0.118	0.006	0.048	0.681	0.116	0.022	0.035	0.161	-0.003	0.093	0.061	0.15	-0.013	0.56	0.825
x26	0.071	0.071	0.093	0.074	0.122	0.741	0.064	0.098	0.036	0.112	0.095	0.075	0.08	-0.054	0.64	0.828
x27	0.153	0.116	0.244	0.081	0.076	0.625	0.063	0.059	0.034	0.159	0.091	0.108	0.05	0.079	0.562	0.858
x25	0.186	0.052	0.072	0.029	0.235	0.29	0.064	0.248	0.062	0.119	0.219	-0.061	0.024	0.122	0.334	0.905
x7	0.1	0.368	0.097	0.047	0.078	0.054	0.733	0.027	0.133	0.055	-0.007	0.088	0.046	0.063	0.739	0.837
x8	0.046	0.369	0.023	-0.014	0.022	0.073	0.707	0.087	0.002	0.01	-0.02	0.051	0.113	-0.045	0.67	0.802
x23	0.103	-0.009	0.03	0.248	0.088	0.128	0.009	0.695	-0.021	0.084	0.188	0.008	0.035	0.028	0.625	0.811
x15	-0.074	-0.017	0.104	0.185	-0.064	-0.003	0.03	0.431	-0.043	-0.015	-0.008	0.101	0.247	0.007	0.315	0.737
x24	0.168	0.098	0.019	0.116	0.104	0.223	0.018	0.353	0.086	0.227	0.159	0.144	-0.069	-0.064	0.35	0.907
x10	0.103	0.037	0.108	-0.027	0.066	0.015	0.04	0.268	0.159	-0.026	0.031	-0.012	0.064	0.001	0.133	0.784
x6	0.076	0.173	0.026	0.024	0.126	0.068	0.127	0.105	0.719	0.059	0.057	0.034	0.041	0.102	0.621	0.8
x5	0.132	0.141	-0.006	0.025	0.188	0.01	0.007	-0.011	0.709	0.039	0.064	0.062	0.011	-0.085	0.594	0.783
x17	0.094	0.091	0.064	0.067	0.065	0.06	0.059	0.002	0.051	0.854	0.033	0.095	-0.02	-0.084	0.787	0.753
x18	0.082	0	0.066	0.019	-0.041	0.158	0.009	0.06	0.02	0.468	0.103	0.024	0.097	0.086	0.289	0.788
x22	0.157	0.07	0.048	0.051	0.113	0.156	0.009	0.247	0.102	0.122	0.732	0.063	0.073	-0.04	0.704	0.843
x20	0.195	0.091	0.047	0.029	0.178	0.156	-0.049	0.088	0.115	0.237	0.364	0.037	0.114	0.247	0.393	0.862
x14	0.174	0.176	0.109	0.042	0.188	0.084	0.076	0.073	0.067	0.107	0.069	0.675	0.052	0.068	0.611	0.895
x1	0.13	0.216	0.038	0.073	0.031	0.171	0.189	0.047	0.07	0.051	0.022	0.301	0.119	-0.121	0.266	0.934
x21	0.133	0.132	-0.011	0.02	0.253	0.013	0.08	0.191	0.066	0.059	0.072	-0.006	0.542	0.003	0.45	0.868
x19	0.037	0.283	0.11	0.073	0.091	0.15	0.062	0.054	0.017	0.033	0.09	0.118	0.388	0.061	0.314	0.9
x16	0	0.102	0.079	0.15	0.257	0.14	0.12	0.1	0.021	0.147	-0.029	0.28	0.313	-0.11	0.36	0.875

Table B1.3

14-Factor, 37 Variables, Quartimax Rotation

	QUARTIMAX															
vars	PA3	PA2	PA4	PA5	PA1	PA10	PA9	PA7	PA6	PA12	PA8	PA11	PA13	PA14	communality	KMO MSAi
x35	0.892	0.066	0.044	0.072	0.058	0.104	-0.008	-0.002	0.02	0.034	-0.023	0.007	-0.009	-0.022	0.824	0.821
x33	0.808	0.097	0.034	-0.008	0.08	-0.056	0.033	0.063	0.011	-0.017	0.056	0.048	-0.085	-0.197	0.731	0.838
x32	0.757	0.126	0.115	0.013	0.152	0.113	-0.029	0.124	0.023	0.079	0.137	0.104	0.004	0.133	0.709	0.894
x36	0.645	0.088	0.1	0.158	-0.019	0.047	0.066	0.015	0.06	0.139	-0.135	-0.078	0.188	0.064	0.553	0.873
x37	0.621	0.074	0.049	0.048	-0.092	-0.039	0.198	0.012	0.131	0.016	0.004	0.002	0.096	0.194	0.509	0.848
x4	0.087	0.882	0.042	0.012	0.071	0.057	0.132	0.052	0.041	0.011	-0.031	0.028	0.017	0.022	0.82	0.838
x3	0.107	0.808	0.032	0.042	0.035	0.016	0.084	0.065	-0.014	0.025	0.002	-0.049	-0.022	0.023	0.685	0.852
x2	0.185	0.701	-0.006	-0.031	0.047	0.014	0.104	0.057	0.079	0.012	-0.007	0.12	0.029	-0.055	0.568	0.908
x11	0.072	0.505	0.081	0.016	-0.017	0.042	0.24	0.148	0.024	0.081	0.181	0.061	0.084	-0.049	0.402	0.906
x9	0.175	0.293	0.055	0.026	0.145	0.067	0.247	0.086	0.062	0.07	-0.094	0.107	-0.061	-0.036	0.248	0.904
x29	0.122	0.058	0.924	0.028	0.041	0.045	0.05	0.02	0.087	-0.031	0.026	-0.026	-0.002	-0.041	0.891	0.682
x28	0.166	0.047	0.631	0.133	0.031	0.129	0.032	-0.005	-0.016	0.054	0.072	0.148	0.011	0.193	0.531	0.762
x30	0.066	0.091	0.484	0.112	-0.041	0.214	0.034	-0.006	-0.003	0.282	-0.1	0.089	0.059	-0.107	0.421	0.815
x34	0.155	0.051	0.017	0.805	0.044	-0.007	0.012	0.023	0.112	0.038	0.045	0.068	-0.034	0.02	0.7	0.693
x31	0.055	0.007	0.106	0.804	0.007	0.134	0.023	0.018	-0.026	0.033	0.102	-0.032	0.06	-0.016	0.697	0.676
x13	0.293	0.127	0.085	0.013	0.69	0.079	0.08	0.183	0.017	0.062	0.05	0.133	0.03	0.025	0.657	0.867
x12	0.136	0.151	0.014	0.058	0.686	0.11	0.013	0.143	0.01	0.091	0.006	0.019	0.056	-0.016	0.56	0.825
x16	0.048	0.135	0.085	0.179	0.291	0.164	0.132	0.011	0.154	0.001	0.054	0.245	0.264	-0.122	0.36	0.875
x26	0.137	0.094	0.087	0.098	0.12	0.74	0.063	0.031	0.085	0.11	0.068	0.027	0.024	-0.047	0.64	0.828
x27	0.208	0.137	0.23	0.103	0.061	0.624	0.059	0.027	0.12	0.102	0.023	0.079	0	0.09	0.562	0.858
x25	0.233	0.063	0.081	0.055	0.217	0.276	0.064	0.061	0.082	0.235	0.216	-0.079	-0.012	0.141	0.334	0.905
x7	0.116	0.4	0.103	0.052	0.071	0.045	0.722	0.121	0.057	0.002	-0.014	0.054	-0.014	0.054	0.739	0.837
x8	0.056	0.396	0.035	-0.001	0.025	0.068	0.701	-0.007	0.017	0	0.056	0.014	0.064	-0.057	0.67	0.802
x6	0.121	0.197	0.037	0.035	0.129	0.06	0.125	0.714	0.044	0.065	0.062	0.02	0.022	0.09	0.621	0.8
x5	0.179	0.157	-0.002	0.024	0.185	-0.002	0.001	0.695	0.051	0.062	-0.057	0.027	-0.007	-0.093	0.594	0.783
x17	0.122	0.092	0.093	0.066	0.028	0.083	0.046	0.048	0.856	0.058	0.013	0.05	-0.003	-0.011	0.787	0.753
x18	0.098	0.008	0.075	0.033	-0.049	0.175	0.01	0.021	0.438	0.123	0.054	0.016	0.108	0.133	0.289	0.788
x22	0.217	0.082	0.018	0.083	0.109	0.15	0.007	0.095	0.091	0.751	0.155	0.041	0.032	-0.009	0.704	0.843
x20	0.232	0.11	0.032	0.046	0.17	0.155	-0.052	0.107	0.194	0.362	0.052	0.035	0.077	0.281	0.393	0.862
x23	0.157	-0.015	0.068	0.31	0.082	0.117	0.033	0.001	0.043	0.249	0.641	0.004	0.042	0.015	0.625	0.811
x15	-0.055	-0.008	0.127	0.237	-0.028	0.014	0.059	-0.028	-0.044	0.05	0.384	0.12	0.261	-0.018	0.315	0.737
x24	0.228	0.094	0.037	0.146	0.079	0.214	0.024	0.091	0.214	0.189	0.321	0.108	-0.075	-0.066	0.35	0.907
x10	0.127	0.039	0.128	0.001	0.065	0.009	0.051	0.161	-0.046	0.063	0.234	-0.012	0.071	-0.008	0.133	0.784
x14	0.242	0.198	0.085	0.064	0.195	0.1	0.089	0.058	0.118	0.072	0.032	0.65	0.006	0.015	0.611	0.895
x1	0.169	0.234	0.028	0.088	0.038	0.175	0.195	0.059	0.062	0.033	0.006	0.261	0.087	-0.143	0.266	0.934
x21	0.156	0.169	0.005	0.061	0.299	0.026	0.099	0.051	0.035	0.107	0.139	-0.02	0.508	0.016	0.45	0.868
x19	0.061	0.316	0.098	0.1	0.124	0.165	0.068	0.003	0.008	0.109	0.01	0.113	0.339	0.059	0.314	0.9

Second Run

Variable X10 was removed due to very low communality. Factors were reduced by one since there is a factor that does not contain any variable.

Table B2.1

13-Factor, 36 Variables, Varimax Rotation

	VARIMAX														
vars	PA3	PA2	PA4	PA1	PA5	PA10	PA7	PA9	PA6	PA8	PA13	PA12	PA11	communality	KMO MSAi
x35	0.872	0.072	0.055	0.122	0.074	0.155	0.013	-0.014	0.026	0.011	0.063	0.095	-0.022	0.828	0.821
x33	0.765	0.113	0.033	0.147	-0.003	-0.014	0.076	0.029	-0.015	0.076	0.002	0.164	-0.117	0.673	0.836
x32	0.722	0.125	0.127	0.208	-0.013	0.152	0.137	-0.043	0.053	0.157	0.102	0.119	0.039	0.693	0.894
x36	0.646	0.079	0.119	0.016	0.153	0.076	0.032	0.064	0.09	-0.07	0.151	-0.027	0.189	0.544	0.876
x37	0.633	0.073	0.056	-0.054	0.036	-0.004	0.043	0.182	0.168	0.033	0.028	-0.007	0.111	0.491	0.849
x4	0.064	0.864	0.05	0.096	0.01	0.076	0.072	0.159	0.042	-0.029	0.014	0.058	0.097	0.814	0.845
x3	0.09	0.805	0.034	0.069	0.055	0.049	0.074	0.113	-0.008	-0.021	0.038	-0.027	0.045	0.691	0.855
x2	0.158	0.692	-0.002	0.074	-0.031	0.024	0.066	0.115	0.07	-0.016	0.018	0.183	0.061	0.571	0.908
x11	0.048	0.49	0.077	0.002	0	0.04	0.137	0.245	0.031	0.165	0.068	0.114	0.106	0.386	0.916
x9	0.147	0.276	0.071	0.166	0.019	0.065	0.091	0.253	0.047	-0.047	0.073	0.163	-0.026	0.244	0.911
x29	0.104	0.056	0.929	0.078	0	0.026	0.028	0.063	0.115	0.068	-0.098	-0.026	-0.013	0.917	0.678
x28	0.15	0.034	0.619	0.043	0.092	0.137	0.016	0.027	0.041	0.114	0.027	0.07	0.072	0.463	0.759
x30	0.033	0.075	0.504	-0.031	0.097	0.193	-0.013	0.042	0.018	-0.032	0.245	0.135	0.061	0.393	0.814
x13	0.225	0.092	0.081	0.71	-0.007	0.085	0.176	0.077	0.007	0.065	0.069	0.172	0.112	0.666	0.866
x12	0.072	0.116	0.004	0.681	0.052	0.119	0.137	0.021	-0.001	0.01	0.096	0.073	0.137	0.552	0.828
x31	0.039	-0.005	0.124	0.01	0.815	0.144	0.015	0.022	-0.016	0.18	0.022	0.001	0.1	0.745	0.674
x34	0.139	0.042	0.042	0.056	0.76	0.008	0.031	0.009	0.111	0.14	0.045	0.104	0.014	0.65	0.695
x26	0.069	0.068	0.105	0.125	0.075	0.702	0.027	0.065	0.111	0.095	0.105	0.122	0.066	0.591	0.828
x27	0.153	0.112	0.247	0.072	0.08	0.647	0.033	0.058	0.162	0.041	0.093	0.115	0.056	0.584	0.858
x25	0.195	0.052	0.085	0.235	0.014	0.299	0.073	0.068	0.119	0.257	0.215	-0.09	0.056	0.34	0.905
x6	0.08	0.16	0.035	0.106	0.018	0.076	0.942	0.115	0.054	0.071	0.05	0.018	0.078	0.969	0.797
x5	0.142	0.157	-0.007	0.243	0.032	0.001	0.532	0.017	0.042	-0.043	0.09	0.092	-0.025	0.409	0.788
x8	0.047	0.363	0.024	0.022	-0.011	0.071	-0.002	0.729	0.007	0.065	-0.021	0.087	0.108	0.696	0.802
x7	0.106	0.372	0.099	0.081	0.048	0.058	0.127	0.704	0.06	0	-0.004	0.103	0.059	0.701	0.839
x17	0.093	0.09	0.065	0.069	0.071	0.055	0.038	0.059	0.819	0.011	0.04	0.128	-0.049	0.729	0.755
x18	0.085	-0.002	0.069	-0.042	0.013	0.157	0.027	0.007	0.483	0.066	0.102	0.015	0.111	**0.3**	0.787
x23	0.118	-0.006	0.056	0.09	0.222	0.12	0.006	0.017	0.068	0.764	0.183	-0.007	0.065	0.715	0.806
x15	-0.065	-0.02	0.121	-0.061	0.178	-0.011	-0.022	0.029	-0.02	0.4	0	0.12	0.269	0.303	0.732
x24	0.173	0.095	0.034	0.108	0.112	0.213	0.082	0.019	0.218	0.33	0.17	0.178	-0.068	0.338	0.909
x22	0.16	0.067	0.058	0.116	0.051	0.15	0.095	0.014	0.115	0.224	0.733	0.079	0.073	0.694	0.842
x20	0.202	0.091	0.052	0.183	0.027	0.17	0.104	-0.048	0.246	0.069	0.363	-0.004	0.142	0.344	0.862
x14	0.185	0.176	0.117	0.2	0.034	0.087	0.067	0.064	0.114	0.069	0.073	0.537	0.064	0.452	0.894
x1	0.129	0.211	0.043	0.036	0.07	0.153	0.06	0.179	0.043	0.04	0.024	0.367	0.089	0.275	0.934
x16	-0.001	0.095	0.086	0.264	0.143	0.122	0.014	0.114	0.142	0.094	-0.03	0.333	0.286	0.357	0.874
x21	0.134	0.129	-0.006	0.264	0.03	0.016	0.051	0.089	0.058	0.141	0.084	0.032	0.511	0.408	0.867
x19	0.039	0.278	0.115	0.097	0.066	0.147	0.014	0.059	0.035	0.038	0.094	0.125	0.407	0.324	0.902

Table B2.2

13-Factor, 36 Variables, Quartimax Rotation

	QUARTIMAX														
vars	PA3	PA2	PA4	PA5	PA1	PA7	PA10	PA9	PA6	PA8	PA13	PA11	PA12	communality	KMO MSAi
x35	0.895	0.066	0.039	0.072	0.061	0.001	0.103	-0.005	0.013	-0.009	0.026	-0.021	0	0.828	0.821
x33	0.787	0.1	0.03	-0.008	0.088	0.061	-0.061	0.036	-0.006	0.051	-0.031	-0.121	0.093	0.673	0.836
x32	0.76	0.124	0.119	0.001	0.159	0.135	0.11	-0.038	0.033	0.14	0.077	0.02	0.041	0.693	0.894
x36	0.647	0.088	0.098	0.157	-0.016	0.031	0.043	0.066	0.056	-0.086	0.138	0.192	-0.097	0.544	0.876
x37	0.624	0.073	0.058	0.039	-0.095	0.048	-0.036	0.187	0.144	0.019	0.017	0.133	-0.067	0.491	0.849
x4	0.088	0.877	0.047	0.007	0.07	0.086	0.057	0.132	0.037	-0.017	0.008	0.026	0.001	0.814	0.845
x3	0.105	0.809	0.029	0.046	0.032	0.083	0.018	0.081	-0.016	-0.01	0.026	-0.019	-0.074	0.691	0.855
x2	0.186	0.702	-0.006	-0.033	0.057	0.071	0.013	0.104	0.079	-0.011	0.009	0.01	0.128	0.571	0.908
x11	0.069	0.503	0.076	0.017	-0.002	0.157	0.033	0.236	0.022	0.157	0.074	0.053	0.088	0.386	0.916
x9	0.179	0.291	0.064	0.018	0.155	0.097	0.059	0.244	0.059	-0.059	0.071	-0.076	0.095	0.244	0.911
x29	0.117	0.058	0.941	0.032	0.039	0.031	0.044	0.046	0.07	0.03	-0.033	-0.011	-0.016	0.917	0.678
x28	0.175	0.045	0.61	0.127	0.03	0.022	0.149	0.026	0.004	0.079	0.068	0.056	0.064	0.463	0.759
x30	0.068	0.093	0.466	0.123	-0.022	-0.009	0.218	0.04	-0.009	-0.061	0.281	0.034	0.119	0.393	0.814
x31	0.055	0.009	0.096	0.835	0.01	0.023	0.123	0.022	-0.026	0.128	0.023	0.044	-0.028	0.745	0.674
x34	0.162	0.05	0.02	0.767	0.047	0.032	-0.011	0.009	0.122	0.089	0.038	-0.031	0.057	0.65	0.695
x13	0.294	0.124	0.088	0.006	0.707	0.182	0.072	0.072	0.019	0.062	0.056	0.016	0.071	0.666	0.866
x12	0.133	0.149	0.007	0.06	0.683	0.145	0.106	0.009	0.008	0.019	0.081	0.04	-0.02	0.552	0.828
x16	0.049	0.137	0.083	0.175	0.313	0.03	0.154	0.139	0.149	0.081	-0.013	0.227	0.276	0.357	0.874
x6	0.111	0.163	0.036	0.029	0.096	0.952	0.045	0.077	0.032	0.036	0.043	0.023	0.002	0.969	0.797
x5	0.178	0.16	-0.012	0.027	0.228	0.528	-0.021	-0.009	0.049	-0.064	0.074	-0.074	0.049	0.409	0.788
x26	0.136	0.094	0.082	0.103	0.129	0.053	0.704	0.068	0.079	0.099	0.099	0.011	0.031	0.591	0.828
x27	0.212	0.134	0.223	0.106	0.063	0.054	0.65	0.057	0.124	0.038	0.097	0.015	0.027	0.584	0.858
x25	0.234	0.061	0.085	0.035	0.204	0.097	0.275	0.056	0.083	0.265	0.205	0.014	-0.154	0.34	0.905
x8	0.053	0.392	0.033	0	0.027	0.033	0.064	0.726	0.009	0.057	-0.009	0.045	0.018	0.696	0.802
x7	0.12	0.4	0.106	0.053	0.073	0.155	0.046	0.69	0.065	-0.019	0.007	-0.008	0.02	0.701	0.839
x17	0.125	0.095	0.093	0.065	0.036	0.053	0.086	0.047	0.82	0.021	0.054	-0.015	0.037	0.729	0.755
x18	0.102	0.007	0.078	0.024	-0.051	0.045	0.177	0.005	0.456	0.076	0.115	0.137	-0.03	0.3	0.787
x23	0.153	-0.01	0.067	0.274	0.07	0.04	0.097	0.027	0.037	0.749	0.18	0.023	-0.007	0.715	0.806
x15	-0.056	-0.004	0.121	0.225	-0.022	-0.001	0.006	0.056	-0.04	0.379	0.024	0.246	0.156	0.303	0.732
x24	0.226	0.096	0.03	0.134	0.087	0.097	0.208	0.021	0.214	0.318	0.162	-0.1	0.123	0.338	0.909
x22	0.218	0.081	0.01	0.075	0.116	0.113	0.15	0.004	0.09	0.219	0.727	0.022	0.027	0.694	0.842
x20	0.241	0.105	0.035	0.039	0.17	0.118	0.166	-0.059	0.217	0.077	0.357	0.116	-0.069	0.344	0.862
x21	0.15	0.169	-0.003	0.063	0.302	0.076	0.019	0.104	0.027	0.151	0.091	0.462	-0.004	0.408	0.867
x19	0.064	0.316	0.099	0.097	0.137	0.034	0.163	0.071	0.008	0.039	0.11	0.356	0.094	0.324	0.902
x14	0.254	0.203	0.098	0.058	0.232	0.067	0.118	0.091	0.143	0.04	0.08	0.011	0.473	0.452	0.894
x1	0.172	0.235	0.025	0.088	0.062	0.068	0.169	0.198	0.058	0.018	0.028	0.044	0.315	0.275	0.934

Third Run

Variable X1 was removed due to very low communality, and the number of factors were reduced to enhance the interpretation.

Table B3.1

12-Factor, 35 Variables, Varimax Rotation

	VARIMAX													
vars	PA3	PA2	PA4	PA1	PA5	PA12	PA10	PA9	PA7	PA6	PA11	PA8	communality	KMO MSAi
x35	0.873	0.078	0.06	0.132	0.074	0.051	0.164	-0.016	0.02	0.033	-0.02	-0.014	0.826	0.819
x33	0.775	0.123	0.033	0.154	0.001	0.091	0.003	0.027	0.086	-0.008	-0.081	-0.175	0.695	0.834
x32	0.72	0.127	0.132	0.207	-0.013	0.199	0.159	-0.04	0.15	0.056	0.063	-0.039	0.693	0.893
x36	0.647	0.081	0.124	0.029	0.144	-0.014	0.078	0.05	0.038	0.101	0.154	0.218	0.554	0.875
x37	0.628	0.074	0.065	-0.05	0.04	0.044	-0.021	0.19	0.038	0.166	0.09	0.087	0.491	0.848
x4	0.062	0.862	0.057	0.099	0.013	-0.021	0.073	0.151	0.077	0.037	0.097	0.026	0.805	0.84
x3	0.085	0.799	0.04	0.059	0.052	0	0.042	0.102	0.079	-0.019	0.033	0.082	0.68	0.853
x2	0.161	0.702	0.003	0.101	-0.026	-0.003	0.036	0.107	0.068	0.085	0.081	-0.063	0.565	0.907
x11	0.048	0.499	0.079	0.001	0	0.177	0.052	0.236	0.149	0.035	0.145	-0.052	0.394	0.913
x9	0.152	0.294	0.07	0.196	0.015	-0.005	0.088	0.238	0.099	0.071	-0.011	-0.009	0.233	0.909
x29	0.094	0.045	0.924	0.057	0.008	0.022	0	0.072	0.028	0.088	-0.016	-0.052	0.885	0.678
x28	0.147	0.034	0.633	0.053	0.094	0.109	0.134	0.032	0.011	0.039	0.079	-0.01	0.474	0.757
x30	0.048	0.094	0.474	0.012	0.078	0.054	0.245	0.014	0	0.064	0.085	0.065	0.321	0.811
x13	0.226	0.098	0.084	0.719	-0.009	0.102	0.09	0.076	0.191	0.01	0.127	-0.034	0.663	0.865
x12	0.071	0.117	0.009	0.686	0.047	0.057	0.113	0.015	0.147	-0.003	0.123	0.072	0.55	0.826
x14	0.206	0.21	0.111	0.273	0.044	0.091	0.158	0.069	0.073	0.169	0.152	-0.259	0.338	0.89
x31	0.033	-0.012	0.13	0.003	0.822	0.168	0.133	0.023	0.016	-0.026	0.103	0.038	0.753	0.672
x34	0.141	0.049	0.042	0.068	0.754	0.149	0.024	0.006	0.032	0.122	0.039	-0.037	0.639	0.696
x23	0.101	-0.019	0.076	0.062	0.231	0.769	0.076	0.028	0.002	0.045	0.103	0.008	0.685	0.805
x22	0.181	0.095	0.045	0.158	0.028	0.407	0.244	-0.045	0.122	0.181	0.111	0.256	0.422	0.84
x24	0.173	0.104	0.036	0.116	0.11	0.392	0.233	0.013	0.091	0.234	-0.017	-0.109	0.351	0.909
x15	-0.068	-0.024	0.131	-0.053	0.19	0.353	-0.03	0.045	-0.033	-0.023	0.308	-0.104	0.296	0.728
x25	0.182	0.044	0.096	0.216	0.006	0.336	0.272	0.066	0.077	0.104	0.03	0.2	0.34	0.906
x26	0.071	0.069	0.115	0.137	0.075	0.143	0.68	0.071	0.032	0.114	0.079	-0.005	0.556	0.824
x27	0.152	0.112	0.254	0.084	0.079	0.083	0.657	0.066	0.035	0.163	0.061	0.002	0.588	0.854
x8	0.048	0.377	0.026	0.03	-0.007	0.051	0.073	0.72	-0.001	0.006	0.124	-0.032	0.689	0.793
x7	0.106	0.387	0.101	0.094	0.05	0.001	0.064	0.703	0.131	0.064	0.067	-0.002	0.706	0.833
x6	0.077	0.163	0.04	0.111	0.02	0.091	0.063	0.122	0.844	0.051	0.07	0.029	0.794	0.795
x5	0.142	0.159	-0.009	0.238	0.027	-0.001	0.02	0.003	0.596	0.054	-0.011	-0.003	0.461	0.785
x17	0.094	0.095	0.073	0.079	0.073	0.038	0.064	0.064	0.044	0.804	-0.034	-0.072	0.698	0.751
x18	0.081	-0.004	0.077	-0.038	0.011	0.098	0.15	0.01	0.028	0.485	0.103	0.072	0.298	0.785
x21	0.132	0.124	0	0.256	0.029	0.142	0.009	0.087	0.059	0.05	0.493	0.09	0.384	0.865
x19	0.047	0.282	0.112	0.107	0.061	0.044	0.173	0.049	0.024	0.046	0.444	0.019	0.343	0.9
x16	0.01	0.108	0.089	0.306	0.153	0.068	0.139	0.123	0.014	0.163	0.334	-0.197	0.353	0.868
x20	0.202	0.1	0.049	0.197	0.003	0.203	0.2	-0.076	0.119	0.275	0.126	0.279	0.363	0.861

Table B3.2

12-Factor, 35 Variables, Quartimax Rotation

	QUARTIMAX													
vars	PA3	PA2	PA4	PA5	PA1	PA10	PA8	PA7	PA9	PA6	PA11	PA12	communality	KMO MSAi
x35	0.894	0.065	0.038	0.072	0.062	0.103	-0.005	-0.007	-0.006	0.011	-0.027	0.009	0.826	0.819
x33	0.797	0.101	0.029	-0.008	0.083	-0.059	0.036	0.057	0.035	-0.004	-0.096	-0.154	0.695	0.834
x32	0.763	0.124	0.115	0.001	0.156	0.119	0.154	0.131	-0.034	0.031	0.034	-0.027	0.693	0.893
x36	0.647	0.091	0.098	0.153	-0.006	0.048	-0.038	0.026	0.056	0.059	0.159	0.241	0.554	0.875
x37	0.621	0.073	0.057	0.041	-0.091	-0.049	0.022	0.031	0.197	0.139	0.104	0.12	0.491	0.848
x4	0.089	0.874	0.045	0.009	0.071	0.054	-0.02	0.069	0.129	0.032	0.023	0.008	0.805	0.84
x3	0.105	0.805	0.024	0.046	0.027	0.018	-0.005	0.071	0.078	-0.024	-0.032	0.062	0.68	0.853
x2	0.186	0.706	-0.002	-0.033	0.069	0.014	-0.007	0.057	0.094	0.087	0.026	-0.072	0.565	0.907
x11	0.073	0.511	0.075	0.014	-0.003	0.048	0.177	0.155	0.228	0.027	0.083	-0.064	0.394	0.913
x9	0.179	0.303	0.069	0.013	0.169	0.072	-0.024	0.095	0.226	0.073	-0.062	-0.001	0.233	0.909
x29	0.118	0.056	0.925	0.036	0.032	0.042	0.013	0.024	0.052	0.062	-0.015	-0.011	0.885	0.678
x28	0.176	0.049	0.62	0.126	0.042	0.158	0.097	0.011	0.025	0.009	0.06	0.01	0.474	0.757
x30	0.077	0.112	0.453	0.106	0.008	0.269	0.049	0.003	0.005	0.029	0.067	0.075	0.321	0.811
x31	0.054	0.007	0.094	0.841	0.009	0.116	0.127	0.019	0.024	-0.032	0.036	0.029	0.753	0.672
x34	0.164	0.056	0.02	0.759	0.052	-0.001	0.104	0.028	0.005	0.131	-0.018	-0.032	0.639	0.696
x13	0.297	0.127	0.085	0.005	0.708	0.074	0.075	0.178	0.073	0.019	0.024	-0.036	0.663	0.865
x12	0.135	0.149	0.004	0.059	0.686	0.103	0.039	0.138	0.008	0.003	0.022	0.061	0.55	0.826
x16	0.053	0.141	0.089	0.178	0.33	0.15	0.07	0.014	0.138	0.165	0.264	-0.194	0.353	0.868
x14	0.257	0.222	0.112	0.058	0.259	0.15	0.072	0.064	0.076	0.175	0.094	-0.25	0.338	0.89
x26	0.138	0.092	0.08	0.106	0.134	0.682	0.118	0.042	0.072	0.079	0.017	-0.012	0.556	0.824
x27	0.214	0.133	0.218	0.109	0.068	0.661	0.055	0.04	0.063	0.122	0.015	0.006	0.588	0.854
x23	0.153	-0.014	0.063	0.276	0.07	0.086	0.752	0.027	0.034	0.029	0.022	-0.008	0.685	0.805
x22	0.232	0.112	0.021	0.059	0.153	0.253	0.397	0.137	-0.052	0.139	0.062	0.248	0.422	0.84
x15	-0.055	-0.005	0.126	0.229	-0.01	-0.008	0.366	-0.017	0.063	-0.034	0.266	-0.118	0.296	0.728
x24	0.231	0.103	0.028	0.129	0.091	0.23	0.362	0.098	0.01	0.227	-0.07	-0.105	0.351	0.909
x25	0.233	0.06	0.076	0.032	0.203	0.275	0.319	0.09	0.058	0.072	-0.026	0.199	0.34	0.906
x6	0.12	0.181	0.038	0.032	0.116	0.05	0.056	0.845	0.096	0.036	0.027	0.011	0.794	0.795
x5	0.182	0.167	-0.009	0.024	0.223	-0.003	-0.037	0.585	-0.019	0.054	-0.049	-0.013	0.461	0.785
x8	0.054	0.401	0.032	0.001	0.028	0.065	0.053	0.017	0.717	0.008	0.047	-0.022	0.689	0.793
x7	0.12	0.411	0.107	0.053	0.078	0.051	-0.012	0.143	0.689	0.066	-0.009	0.015	0.706	0.833
x17	0.129	0.095	0.091	0.065	0.037	0.093	0.032	0.051	0.05	0.802	-0.014	-0.011	0.698	0.751
x18	0.102	0.006	0.075	0.023	-0.046	0.178	0.105	0.04	0.008	0.457	0.122	0.103	0.298	0.785
x21	0.154	0.166	-0.008	0.064	0.3	0.019	0.165	0.064	0.107	0.024	0.435	0.077	0.384	0.865
x19	0.07	0.32	0.094	0.094	0.143	0.186	0.064	0.026	0.063	0.015	0.393	0.003	0.343	0.9
x20	0.244	0.12	0.03	0.024	0.187	0.211	0.199	0.127	-0.085	0.234	0.1	0.284	0.363	0.861

Fourth Run

Variable X15 was removed due to its low communality and factor was reduced once again by one since there is a factor that does not contain any variable.

Table B4.1

11-Factor, 34 Variables, Varimax Rotation

VARIMAX													
vars	PA3	PA2	PA1	PA4	PA9	PA5	PA8	PA7	PA10	PA6	PA11	communality	KMO MSAi
x35	0.872	0.076	0.126	0.061	0.109	0.068	-0.027	0.029	0.15	0.027	-0.033	0.829	0.817
x33	0.771	0.115	0.148	0.034	0.104	0.006	0.032	0.085	-0.02	-0.015	-0.2	0.691	0.834
x32	0.702	0.125	0.223	0.143	0.275	-0.003	-0.03	0.136	0.103	0.039	-0.029	0.686	0.896
x36	0.646	0.088	0.044	0.122	0.053	0.135	0.033	0.055	0.084	0.102	0.279	0.563	0.874
x37	0.625	0.08	-0.023	0.064	0.07	0.043	0.188	0.035	-0.021	0.171	0.135	0.494	0.848
x4	0.062	0.866	0.126	0.056	0.004	0.008	0.133	0.086	0.067	0.035	0.054	0.807	0.84
x3	0.089	0.793	0.053	0.035	0.021	0.04	0.082	0.1	0.046	-0.025	0.049	0.665	0.853
x2	0.159	0.706	0.141	0.003	0.011	-0.024	0.1	0.064	0.025	0.087	-0.016	0.567	0.907
x11	0.034	0.504	0.062	0.087	0.182	0.023	0.258	0.123	0.016	0.032	0.018	0.383	0.911
x19	0.024	0.3	0.252	0.128	0.111	0.084	0.083	-0.023	0.123	0.053	0.261	0.283	0.898
x9	0.158	0.29	0.184	0.066	0.007	0	0.212	0.131	0.093	0.066	-0.035	0.223	0.909
x13	0.225	0.087	0.708	0.076	0.137	-0.029	0.043	0.234	0.075	-0.022	-0.022	0.649	0.864
x12	0.076	0.107	0.635	0	0.11	0.02	-0.023	0.199	0.109	-0.034	0.06	0.49	0.827
x16	-0.004	0.124	0.438	0.101	0.086	0.169	0.142	-0.022	0.097	0.161	0.039	0.311	0.869
x21	0.105	0.143	0.385	0.012	0.197	0.054	0.114	0.02	-0.032	0.038	0.33	0.346	0.862
x14	0.194	0.215	0.36	0.123	0.108	0.057	0.084	0.047	0.108	0.168	-0.12	0.307	0.892
x29	0.099	0.043	0.062	0.892	-0.007	0.009	0.066	0.039	0.006	0.088	-0.06	0.828	0.677
x28	0.138	0.037	0.087	0.655	0.12	0.099	0.036	0.001	0.095	0.037	0.02	0.493	0.752
x30	0.041	0.099	0.034	0.487	0.104	0.076	0.014	0.001	0.216	0.062	0.088	0.324	0.81
x23	0.077	-0.023	0.091	0.091	0.666	0.274	0.061	-0.022	0.014	0.021	-0.022	0.547	0.819
x22	0.156	0.091	0.129	0.058	0.528	0.03	-0.043	0.128	0.169	0.139	0.181	0.431	0.837
x24	0.151	0.1	0.125	0.05	0.452	0.13	0.036	0.066	0.15	0.212	-0.182	0.378	0.907
x25	0.167	0.037	0.173	0.104	0.418	0.007	0.054	0.095	0.22	0.068	0.092	0.318	0.914
x20	0.187	0.098	0.176	0.055	0.326	-0.008	-0.09	0.137	0.151	0.243	0.225	0.344	0.86
x31	0.025	-0.008	0.04	0.136	0.15	0.841	0.025	0.014	0.122	-0.03	0.075	0.772	0.653
x34	0.135	0.047	0.095	0.046	0.134	0.748	0.004	0.032	0.006	0.117	-0.03	0.624	0.687
x8	0.044	0.383	0.09	0.031	0.046	0.003	0.749	-0.01	0.06	0.005	0.037	0.726	0.795
x7	0.11	0.396	0.125	0.102	-0.006	0.045	0.663	0.148	0.067	0.068	0.022	0.668	0.833
x6	0.071	0.172	0.126	0.041	0.138	0.024	0.126	0.75	0.041	0.05	0.043	0.656	0.795
x5	0.135	0.148	0.2	-0.012	0.042	0.021	-0.006	0.654	0.008	0.042	-0.025	0.513	0.785
x26	0.063	0.074	0.171	0.119	0.22	0.074	0.068	0.029	0.691	0.111	0.007	0.602	0.825
x27	0.146	0.121	0.123	0.267	0.176	0.077	0.06	0.027	0.616	0.168	0.018	0.571	0.856
x17	0.095	0.091	0.098	0.071	0.079	0.067	0.051	0.055	0.052	0.776	-0.085	0.66	0.76
x18	0.072	0.002	0.001	0.079	0.15	0.015	0.013	0.017	0.129	0.495	0.109	0.308	0.787

Table B4.2

11-Factor, 34 Variables, Quartimax Rotation

QUARTIMAX													
vars	PA1	PA2	PA4	PA5	PA9	PA10	PA8	PA7	PA3	PA6	PA11	communality	KMO MSAi
x35	0.897	0.066	0.036	0.068	0.05	0.107	-0.014	-0.006	-0.01	0.008	-0.009	0.829	0.817
x33	0.796	0.1	0.028	-0.003	0.083	-0.061	0.04	0.051	0.019	-0.005	-0.179	0.691	0.834
x32	0.757	0.126	0.123	0.015	0.166	0.101	-0.022	0.111	0.172	0.028	-0.009	0.686	0.896
x36	0.65	0.094	0.093	0.144	-0.017	0.061	0.044	0.034	-0.033	0.057	0.301	0.563	0.874
x37	0.622	0.075	0.053	0.041	-0.087	-0.04	0.194	0.022	0.01	0.144	0.168	0.494	0.848
x4	0.09	0.876	0.044	0.007	0.067	0.054	0.13	0.061	-0.012	0.028	0.018	0.807	0.84
x3	0.11	0.798	0.019	0.038	0.003	0.028	0.077	0.077	0.004	-0.032	0.012	0.665	0.853
x2	0.185	0.71	-0.002	-0.03	0.083	0.011	0.099	0.039	-0.01	0.09	-0.036	0.567	0.907
x11	0.07	0.511	0.08	0.034	0.028	0.034	0.25	0.122	0.171	0.03	-0.005	0.383	0.911
x19	0.059	0.327	0.108	0.112	0.225	0.147	0.091	-0.031	0.082	0.033	0.243	0.283	0.898
x9	0.185	0.301	0.065	0.004	0.144	0.085	0.213	0.115	-0.032	0.067	-0.044	0.223	0.909
x29	0.124	0.056	0.893	0.038	0.028	0.047	0.052	0.026	-0.008	0.062	-0.022	0.828	0.677
x28	0.175	0.05	0.64	0.133	0.056	0.134	0.029	-0.006	0.092	0.01	0.04	0.493	0.752
x30	0.074	0.114	0.466	0.106	0.009	0.25	0.007	-0.001	0.07	0.027	0.096	0.324	0.81
x31	0.056	0.006	0.095	0.859	0.01	0.115	0.023	0.017	0.077	-0.033	0.042	0.772	0.653
x34	0.166	0.055	0.021	0.753	0.051	-0.004	0.003	0.027	0.065	0.13	-0.043	0.624	0.687
x13	0.306	0.127	0.079	0.002	0.689	0.092	0.062	0.202	0.065	0.007	-0.036	0.649	0.864
x12	0.147	0.147	-0.003	0.05	0.628	0.126	-0.005	0.173	0.048	-0.012	0.036	0.49	0.827
x16	0.047	0.153	0.099	0.193	0.405	0.123	0.154	-0.033	0.048	0.175	0.032	0.311	0.869
x21	0.145	0.175	-0.001	0.087	0.372	0.003	0.129	0.015	0.172	0.031	0.318	0.346	0.862
x14	0.25	0.23	0.124	0.07	0.313	0.121	0.091	0.027	0.053	0.184	-0.116	0.307	0.892
x26	0.14	0.095	0.08	0.112	0.134	0.717	0.071	0.031	0.102	0.077	-0.011	0.602	0.825
x27	0.213	0.139	0.231	0.112	0.073	0.641	0.06	0.024	0.065	0.126	0.015	0.571	0.856
x8	0.054	0.393	0.034	0.009	0.039	0.058	0.748	-0.003	0.045	0.005	0.009	0.726	0.795
x7	0.126	0.411	0.106	0.048	0.068	0.059	0.66	0.145	-0.029	0.065	0.001	0.668	0.833
x6	0.124	0.2	0.042	0.037	0.131	0.051	0.11	0.746	0.076	0.044	0.027	0.656	0.795
x5	0.18	0.171	-0.007	0.024	0.201	0	-0.015	0.637	-0.021	0.048	-0.035	0.513	0.785
x23	0.161	-0.02	0.066	0.32	0.088	0.094	0.055	0.008	0.627	0.031	-0.031	0.547	0.819
x22	0.235	0.106	0.029	0.075	0.121	0.24	-0.045	0.145	0.464	0.117	0.182	0.431	0.837
x24	0.231	0.099	0.039	0.155	0.095	0.203	0.03	0.077	0.388	0.224	-0.174	0.378	0.907
x25	0.239	0.053	0.08	0.048	0.16	0.274	0.054	0.106	0.352	0.051	0.09	0.318	0.914
x20	0.247	0.117	0.034	0.025	0.158	0.204	-0.087	0.14	0.264	0.216	0.24	0.344	0.86
x17	0.133	0.097	0.089	0.064	0.036	0.099	0.045	0.056	0.042	0.776	-0.019	0.66	0.76
x18	0.102	0.009	0.077	0.028	-0.032	0.177	0.01	0.028	0.115	0.47	0.153	0.308	0.787

Fifth Run

Variable X19 was removed due to low communality and the factor was reduced by one for better interpretation.

Table B5.1

10-Factor, 33 Variables, Varimax Rotation

VARIMAX												
vars	PA3	PA2	PA1	PA9	PA4	PA5	PA8	PA7	PA6	PA10	communality	KMO MSAi
x35	0.884	0.068	0.126	0.113	0.055	0.065	-0.032	0.022	0.023	0.15	0.847	0.818
x33	0.764	0.1	0.15	0.065	0.032	0.01	0.025	0.091	-0.004	-0.012	0.631	0.833
x32	0.709	0.109	0.216	0.277	0.138	-0.001	-0.027	0.135	0.035	0.093	0.686	0.896
x37	0.619	0.083	-0.019	0.107	0.06	0.037	0.185	0.028	0.162	-0.028	0.469	0.847
x36	0.614	0.1	0.053	0.137	0.117	0.119	0.037	0.039	0.087	0.071	0.452	0.873
x4	0.065	0.862	0.135	0.032	0.059	0.007	0.141	0.079	0.035	0.066	0.802	0.837
x3	0.09	0.798	0.064	0.045	0.038	0.041	0.086	0.089	-0.027	0.044	0.672	0.846
x2	0.162	0.705	0.149	0.02	0.005	-0.023	0.102	0.058	0.09	0.029	0.57	0.905
x11	0.041	0.491	0.061	0.18	0.087	0.028	0.266	0.126	0.032	0.015	0.375	0.907
x9	0.158	0.292	0.201	0.016	0.069	0.001	0.207	0.115	0.068	0.09	0.224	0.913
x13	0.223	0.079	0.726	0.153	0.077	-0.029	0.039	0.211	-0.022	0.065	0.664	0.861
x12	0.066	0.11	0.659	0.149	0.002	0.019	-0.028	0.168	-0.038	0.099	0.513	0.825
x16	0	0.113	0.418	0.101	0.1	0.17	0.146	-0.015	0.162	0.101	0.294	0.857
x14	0.202	0.196	0.347	0.095	0.122	0.06	0.085	0.055	0.172	0.108	0.279	0.889
x21	0.105	0.14	0.337	0.25	0.009	0.055	0.124	0.03	0.029	-0.02	0.227	0.879
x23	0.081	-0.036	0.088	0.608	0.091	0.287	0.066	-0.025	0.017	0.008	0.481	0.82
x22	0.147	0.087	0.122	0.592	0.057	0.032	-0.028	0.105	0.12	0.131	0.443	0.833
x25	0.161	0.035	0.175	0.453	0.105	0.011	0.059	0.073	0.054	0.197	0.325	0.913
x20	0.179	0.101	0.172	0.398	0.054	-0.008	-0.075	0.111	0.222	0.123	0.315	0.857
x24	0.159	0.079	0.124	0.392	0.055	0.142	0.038	0.068	0.204	0.142	0.291	0.906
x29	0.105	0.039	0.067	-0.018	0.893	0.011	0.06	0.037	0.093	0.007	0.828	0.674
x28	0.143	0.031	0.083	0.127	0.655	0.101	0.038	-0.001	0.035	0.089	0.494	0.748
x30	0.042	0.094	0.026	0.137	0.483	0.073	0.02	-0.001	0.058	0.202	0.314	0.804
x31	0.029	-0.01	0.035	0.16	0.135	0.824	0.028	0.011	-0.031	0.118	0.741	0.646
x34	0.14	0.042	0.095	0.12	0.044	0.764	0.002	0.03	0.118	0.004	0.644	0.684
x8	0.049	0.376	0.097	0.045	0.033	0.006	0.761	-0.012	0.007	0.061	0.739	0.791
x7	0.116	0.392	0.137	0	0.104	0.044	0.658	0.143	0.071	0.066	0.661	0.83
x6	0.073	0.17	0.129	0.16	0.04	0.023	0.126	0.766	0.046	0.038	0.685	0.795
x5	0.139	0.147	0.21	0.053	-0.013	0.017	-0.01	0.646	0.043	0.009	0.508	0.783
x17	0.097	0.089	0.101	0.087	0.071	0.066	0.043	0.049	0.774	0.052	0.65	0.766
x18	0.073	-0.003	-0.01	0.189	0.076	0.011	0.018	0.018	0.486	0.117	0.297	0.801
x26	0.063	0.065	0.164	0.248	0.121	0.074	0.067	0.027	0.111	0.683	0.601	0.82
x27	0.147	0.114	0.116	0.205	0.269	0.076	0.059	0.027	0.167	0.618	0.582	0.852
x17	0.099	0.086	0.105	0.094	0.056	0.064	0.05	0.075	0.083	0.759	0.635	0.801
x18	0.078	0.01	0.005	0.085	0.136	0.012	0.017	0.134	0.021	0.458	0.261	0.816

Table B5.2

10-Factor, 33 Variables, Quartimax Rotation

	QUARTIMAX											
vars	PA1	PA2	PA4	PA5	PA9	PA3	PA8	PA7	PA10	PA6	communality	KMO MSAi
x35	0.907	0.067	0.035	0.069	0.052	0.106	-0.014	-0.006	-0.018	0.005	0.847	0.818
x33	0.776	0.099	0.025	0.009	0.089	-0.054	0.04	0.063	-0.027	-0.011	0.631	0.833
x32	0.761	0.118	0.121	0.023	0.165	0.095	-0.015	0.119	0.162	0.026	0.686	0.896
x36	0.635	0.099	0.102	0.126	-0.007	0.054	0.045	0.024	0.046	0.071	0.452	0.873
x37	0.623	0.076	0.057	0.036	-0.084	-0.045	0.19	0.021	0.044	0.149	0.469	0.847
x4	0.089	0.871	0.046	0.007	0.072	0.055	0.142	0.064	-0.001	0.031	0.802	0.837
x3	0.109	0.802	0.022	0.039	0.011	0.029	0.084	0.074	0.014	-0.033	0.672	0.846
x2	0.183	0.712	-0.001	-0.026	0.088	0.015	0.106	0.041	-0.015	0.09	0.57	0.905
x11	0.071	0.499	0.078	0.042	0.025	0.036	0.26	0.133	0.159	0.029	0.375	0.907
x9	0.18	0.305	0.067	0.007	0.157	0.083	0.213	0.105	-0.033	0.066	0.224	0.913
x29	0.122	0.056	0.894	0.037	0.031	0.05	0.051	0.025	-0.017	0.061	0.828	0.674
x28	0.176	0.046	0.642	0.134	0.054	0.133	0.033	-0.005	0.097	0.009	0.494	0.748
x30	0.078	0.107	0.465	0.101	0.004	0.242	0.014	0.001	0.098	0.031	0.314	0.804
x31	0.058	0.002	0.099	0.841	0.006	0.119	0.025	0.017	0.071	-0.035	0.741	0.646
x34	0.164	0.053	0.022	0.771	0.05	-0.002	0.004	0.028	0.034	0.124	0.644	0.684
x13	0.3	0.123	0.08	0.007	0.708	0.082	0.068	0.188	0.061	0.009	0.664	0.861
x12	0.14	0.151	0.001	0.052	0.653	0.119	-0.003	0.151	0.068	-0.008	0.513	0.825
x16	0.049	0.142	0.099	0.194	0.385	0.128	0.162	-0.018	0.045	0.177	0.294	0.857
x21	0.157	0.16	0.006	0.083	0.322	0.015	0.136	0.034	0.21	0.049	0.227	0.879
x14	0.248	0.219	0.12	0.079	0.302	0.122	0.099	0.043	0.025	0.177	0.279	0.889
x26	0.141	0.089	0.08	0.111	0.135	0.716	0.072	0.035	0.104	0.076	0.601	0.82
x27	0.216	0.134	0.231	0.11	0.072	0.649	0.061	0.03	0.07	0.126	0.582	0.852
x8	0.053	0.383	0.034	0.011	0.037	0.059	0.763	0.003	0.036	0.004	0.739	0.791
x7	0.126	0.404	0.107	0.047	0.07	0.057	0.658	0.147	-0.033	0.065	0.661	0.83
x6	0.124	0.192	0.041	0.037	0.13	0.052	0.108	0.768	0.081	0.041	0.685	0.795
x5	0.179	0.168	-0.008	0.023	0.206	0.001	-0.017	0.634	-0.024	0.046	0.508	0.783
x23	0.157	-0.024	0.064	0.34	0.095	0.099	0.061	0.01	0.559	0.027	0.481	0.82
x22	0.237	0.102	0.033	0.083	0.125	0.221	-0.032	0.132	0.52	0.12	0.443	0.833
x25	0.238	0.052	0.083	0.057	0.171	0.263	0.06	0.091	0.378	0.051	0.325	0.913
x20	0.251	0.116	0.041	0.028	0.16	0.186	-0.074	0.123	0.326	0.222	0.315	0.857
x24	0.225	0.092	0.037	0.177	0.103	0.2	0.038	0.085	0.316	0.201	0.291	0.906
x17	0.131	0.099	0.089	0.071	0.039	0.103	0.045	0.055	0.038	0.769	0.65	0.766
x18	0.109	0.001	0.078	0.026	-0.041	0.171	0.015	0.033	0.147	0.475	0.297	0.801

Sixth Run

Variables X31 and X34 were removed due to low KMO MSAi. Factors was reduced once again by one to accommodate the change brought by the omission.

Table B6.1

9-Factor, 31 Variables, Varimax Rotation

VARIMAX											
vars	PA3	PA2	PA8	PA1	PA4	PA7	PA6	PA5	PA9	communality	KMO MSAi
x35	0.885	0.068	0.122	0.126	0.057	-0.03	0.022	0.025	0.152	0.847	0.819
x33	0.764	0.098	0.07	0.148	0.033	0.026	0.092	-0.006	-0.011	0.63	0.831
x32	0.708	0.109	0.273	0.213	0.136	-0.03	0.142	0.03	0.096	0.682	0.9
x37	0.62	0.081	0.112	-0.018	0.061	0.185	0.03	0.162	-0.028	0.47	0.844
x36	0.613	0.099	0.154	0.056	0.122	0.042	0.036	0.09	0.072	0.444	0.868
x4	0.066	0.862	0.03	0.138	0.059	0.147	0.08	0.041	0.064	0.805	0.84
x3	0.091	0.795	0.046	0.068	0.04	0.095	0.089	-0.022	0.045	0.668	0.846
x2	0.162	0.702	0.016	0.149	0.005	0.109	0.063	0.085	0.034	0.566	0.915
x11	0.04	0.489	0.186	0.06	0.088	0.269	0.129	0.029	0.015	0.377	0.905
x9	0.156	0.288	0.015	0.197	0.071	0.212	0.119	0.056	0.101	0.224	0.945
x23	0.084	-0.035	0.634	0.089	0.101	0.065	-0.026	0.026	0.015	0.434	0.831
x22	0.147	0.09	0.584	0.121	0.053	-0.037	0.113	0.114	0.134	0.433	0.837
x25	0.16	0.036	0.448	0.174	0.101	0.051	0.079	0.051	0.195	0.318	0.92
x24	0.157	0.076	0.424	0.118	0.062	0.043	0.068	0.194	0.146	0.293	0.904
x20	0.179	0.103	0.377	0.171	0.048	-0.081	0.122	0.211	0.132	0.3	0.856
x13	0.223	0.078	0.14	0.724	0.071	0.036	0.218	-0.024	0.07	0.659	0.861
x12	0.066	0.109	0.143	0.665	-0.001	-0.03	0.171	-0.036	0.1	0.52	0.825
x16	0.002	0.11	0.145	0.41	0.109	0.155	-0.019	0.16	0.096	0.273	0.852
x14	0.201	0.192	0.115	0.341	0.127	0.093	0.056	0.161	0.114	0.274	0.884
x21	0.105	0.139	0.263	0.339	0.009	0.123	0.03	0.031	-0.027	0.232	0.877
x29	0.106	0.038	-0.017	0.07	0.885	0.059	0.039	0.095	0.007	0.816	0.68
x28	0.142	0.028	0.145	0.082	0.662	0.041	-0.001	0.032	0.091	0.498	0.74
x30	0.042	0.093	0.147	0.027	0.486	0.022	-0.001	0.056	0.203	0.314	0.798
x8	0.048	0.37	0.052	0.098	0.033	0.755	-0.01	0.005	0.058	0.727	0.792
x7	0.116	0.384	0.008	0.138	0.104	0.667	0.143	0.071	0.067	0.666	0.834
x6	0.071	0.166	0.159	0.125	0.04	0.128	0.774	0.044	0.039	0.694	0.795
x5	0.139	0.145	0.053	0.209	-0.013	-0.007	0.639	0.045	0.008	0.497	0.784
x17	0.098	0.086	0.099	0.105	0.073	0.047	0.047	0.789	0.052	0.672	0.778
x18	0.074	-0.002	0.185	-0.008	0.073	0.014	0.021	0.483	0.118	0.294	0.799
x26	0.063	0.064	0.26	0.164	0.123	0.068	0.026	0.111	0.671	0.586	0.816
x27	0.146	0.111	0.215	0.115	0.271	0.062	0.027	0.164	0.627	0.59	0.848

Table B6.2

9-Factor, 31 Variables, Quartimax Rotation

						QUARTIMAX					
vars	PA1	PA2	PA5	PA4	PA3	PA7	PA9	PA6	PA8	communality	KMO MSAi
x35	0.909	0.066	0.037	0.054	0.108	-0.013	-0.01	-0.008	0.009	0.847	0.819
x33	0.776	0.099	0.025	0.087	-0.055	0.041	-0.025	0.062	-0.011	0.63	0.831
x32	0.761	0.119	0.119	0.16	0.095	-0.019	0.155	0.123	0.024	0.682	0.9
x36	0.637	0.099	0.109	-0.002	0.06	0.048	0.067	0.022	0.076	0.444	0.868
x37	0.625	0.077	0.057	-0.085	-0.044	0.189	0.045	0.022	0.151	0.47	0.844
x4	0.09	0.874	0.046	0.073	0.052	0.136	-0.002	0.062	0.036	0.805	0.84
x3	0.11	0.801	0.024	0.013	0.031	0.082	0.018	0.074	-0.029	0.668	0.846
x2	0.181	0.711	-0.003	0.085	0.017	0.104	-0.023	0.043	0.084	0.566	0.915
x11	0.072	0.501	0.079	0.021	0.038	0.256	0.165	0.136	0.027	0.377	0.905
x9	0.179	0.305	0.068	0.154	0.093	0.214	-0.034	0.108	0.054	0.224	0.945
x29	0.124	0.057	0.887	0.031	0.051	0.05	-0.019	0.027	0.066	0.816	0.68
x28	0.179	0.045	0.652	0.053	0.142	0.036	0.12	-0.003	0.009	0.498	0.74
x30	0.081	0.106	0.469	0.003	0.249	0.014	0.112	0.002	0.031	0.314	0.798
x13	0.3	0.125	0.078	0.705	0.083	0.066	0.055	0.19	0.01	0.659	0.861
x12	0.142	0.152	0.002	0.66	0.118	-0.004	0.074	0.149	-0.002	0.52	0.825
x16	0.056	0.141	0.113	0.382	0.136	0.17	0.107	-0.021	0.178	0.273	0.852
x21	0.16	0.161	0.01	0.322	0.013	0.135	0.233	0.033	0.055	0.232	0.877
x14	0.249	0.217	0.126	0.298	0.131	0.104	0.05	0.044	0.167	0.274	0.884
x26	0.144	0.089	0.084	0.137	0.709	0.072	0.123	0.036	0.081	0.586	0.816
x27	0.218	0.133	0.234	0.072	0.662	0.062	0.081	0.031	0.126	0.59	0.848
x8	0.054	0.388	0.033	0.034	0.059	0.752	0.041	0.005	0.005	0.727	0.792
x7	0.127	0.406	0.107	0.071	0.061	0.661	-0.025	0.147	0.066	0.666	0.834
x23	0.169	-0.023	0.084	0.097	0.122	0.064	0.607	0.012	0.04	0.434	0.831
x22	0.24	0.104	0.031	0.118	0.224	-0.041	0.514	0.14	0.119	0.433	0.837
x25	0.24	0.055	0.081	0.165	0.262	0.053	0.374	0.097	0.053	0.318	0.92
x24	0.229	0.091	0.048	0.099	0.214	0.042	0.357	0.087	0.195	0.293	0.904
x20	0.252	0.118	0.036	0.154	0.191	-0.081	0.303	0.131	0.214	0.3	0.856
x6	0.124	0.192	0.042	0.13	0.053	0.107	0.081	0.775	0.041	0.694	0.795
x5	0.18	0.168	-0.007	0.21	0	-0.016	-0.021	0.624	0.048	0.497	0.784
x17	0.132	0.099	0.091	0.041	0.105	0.045	0.047	0.054	0.785	0.672	0.778
x18	0.11	0.004	0.074	-0.042	0.17	0.009	0.139	0.036	0.474	0.294	0.799

Table B6.3

9-Factor, 31 Variables, Oblimin Rotation

	OBLIMIN										
vars	PA1	PA2	PA5	PA4	PA7	PA3	PA9	PA6	PA8	communality	KMO MSAi
x35	0.918	0.005	-0.023	0.014	-0.038	-0.033	0.099	-0.039	-0.021	0.847	0.819
x33	0.786	0.019	-0.004	0.062	0.024	-0.031	-0.083	0.033	-0.027	0.63	0.831
x32	0.677	0.048	0.067	0.098	-0.066	0.143	0.027	0.079	-0.019	0.682	0.9
x37	0.633	-0.051	0.014	-0.126	0.195	0.046	-0.09	0.01	0.138	0.47	0.844
x36	0.611	0.032	0.064	-0.049	0.023	0.058	0.02	-0.001	0.049	0.444	0.868
x4	-0.036	0.877	0.015	0.02	0.035	-0.016	0.011	-0.008	0.014	0.805	0.84
x3	0.012	0.834	0.003	-0.047	-0.016	0.016	0	0.022	-0.054	0.668	0.846
x2	0.084	0.707	-0.035	0.05	0.019	-0.038	-0.022	-0.023	0.074	0.566	0.915
x11	-0.047	0.4	0.047	-0.048	0.22	0.172	-0.028	0.102	-0.009	0.377	0.905
x29	-0.008	-0.006	0.915	0.009	0	-0.048	-0.032	0.011	0.024	0.816	0.68
x28	0.049	-0.004	0.651	0.017	-0.004	0.104	0.064	-0.028	-0.044	0.498	0.74
x30	-0.04	0.081	0.445	-0.05	-0.03	0.086	0.206	-0.012	-0.021	0.314	0.798
x13	0.101	-0.036	0.043	0.738	0.04	0.002	-0.002	0.071	-0.008	0.659	0.861
x12	-0.051	0.054	-0.031	0.693	-0.039	0.026	0.049	0.035	-0.016	0.52	0.825
x16	-0.106	0.01	0.065	0.407	0.163	0.071	0.056	-0.116	0.167	0.273	0.852
x21	0.023	0.066	-0.019	0.326	0.123	0.239	-0.089	-0.049	0.037	0.232	0.877
x14	0.11	0.111	0.075	0.293	0.071	0.008	0.067	-0.033	0.152	0.274	0.884
x8	-0.01	0.038	-0.028	0.012	0.837	0.032	0.031	-0.036	-0.029	0.727	0.792
x7	0.038	0.06	0.048	0.029	0.715	-0.056	0.033	0.116	0.033	0.666	0.834
x9	0.093	0.169	0.027	0.129	0.2	-0.07	0.073	0.068	0.034	0.224	0.945
x23	0.007	-0.064	0.042	0.03	0.05	0.662	-0.03	-0.035	-0.018	0.434	0.831
x22	0.054	0.085	-0.034	0.02	-0.098	0.533	0.099	0.099	0.061	0.433	0.837
x25	0.08	-0.024	0.014	0.092	0.026	0.371	0.176	0.055	-0.005	0.318	0.92
x24	0.076	0.027	-0.022	0.026	0.005	0.354	0.117	0.047	0.154	0.293	0.904
x20	0.093	0.103	-0.026	0.086	-0.146	0.29	0.097	0.086	0.183	0.3	0.856
x26	-0.013	-0.024	-0.051	0.049	0.049	0.03	0.748	0.01	0.003	0.586	0.816
x27	0.063	0.03	0.109	-0.022	0.023	-0.013	0.688	0.009	0.049	0.59	0.848
x6	-0.034	-0.013	0.012	-0.023	0.06	0.038	0.019	0.827	-0.003	0.694	0.795
x5	0.058	0.038	-0.025	0.111	-0.072	-0.068	-0.022	0.649	0.028	0.497	0.784
x17	-0.011	0.012	0.008	0.016	-0.012	-0.016	0.001	0.006	0.821	0.672	0.778
x18	0.008	-0.044	0.006	-0.088	-0.027	0.105	0.103	0.02	0.476	0.294	0.799
x27	0.025	0.039	0.312	0.095	0.01	0.036	0.18	-0.037	0.339	0.442	0.866
x26	-0.055	0.005	0.186	0.172	0.014	0.06	0.235	-0.043	0.325	0.37	0.845

Seventh Run

Variable X9 and X14 was removed due to low communality, the factors are now reduced to eight (8) for better interpretability.

Table B7.1

8-Factor, 29 Variables, Varimax Rotation

VARIMAX										
vars	PA3	PA2	PA1	PA5	PA4	PA7	PA6	PA8	communality	KMO MSAi
x35	0.869	0.071	0.2	0.078	0.12	-0.035	0.005	0.029	0.823	0.813
x33	0.768	0.091	0.079	0.031	0.124	0.024	0.102	-0.017	0.632	0.823
x32	0.707	0.103	0.313	0.136	0.183	-0.028	0.149	0.012	0.683	0.897
x37	0.624	0.072	0.099	0.052	-0.032	0.179	0.041	0.158	0.467	0.838
x36	0.613	0.095	0.18	0.126	0.061	0.035	0.026	0.096	0.448	0.867
x4	0.071	0.865	0.056	0.072	0.137	0.154	0.081	0.049	0.813	0.828
x3	0.095	0.798	0.058	0.049	0.075	0.103	0.088	-0.016	0.676	0.838
x2	0.167	0.695	0.041	0.013	0.13	0.117	0.073	0.085	0.556	0.914
x11	0.046	0.474	0.18	0.08	0.033	0.272	0.151	0.014	0.363	0.901
x22	0.141	0.077	0.606	0.034	0.087	-0.028	0.124	0.073	0.423	0.832
x23	0.095	-0.048	0.542	0.066	0.062	0.072	0.017	-0.001	0.318	0.828
x25	0.153	0.033	0.501	0.104	0.173	0.05	0.067	0.043	0.325	0.921
x26	0.043	0.095	0.485	0.204	0.18	0.052	-0.035	0.139	0.344	0.809
x24	0.152	0.07	0.469	0.056	0.079	0.052	0.082	0.162	0.294	0.898
x27	0.119	0.134	0.45	0.331	0.132	0.054	-0.032	0.18	0.398	0.842
x20	0.173	0.096	0.426	0.042	0.15	-0.079	0.119	0.195	0.303	0.853
x29	0.113	0.026	-0.01	0.862	0.062	0.058	0.051	0.099	0.777	0.691
x28	0.141	0.018	0.175	0.658	0.053	0.045	0.013	0.016	0.489	0.748
x30	0.03	0.092	0.234	0.507	0.006	0.018	-0.011	0.045	0.324	0.789
x12	0.066	0.11	0.179	0.012	0.715	-0.029	0.154	-0.018	0.584	0.812
x13	0.228	0.078	0.186	0.081	0.689	0.04	0.219	-0.016	0.624	0.857
x16	0.006	0.106	0.195	0.116	0.364	0.158	-0.002	0.149	0.242	0.862
x21	0.117	0.128	0.227	-0.006	0.333	0.139	0.051	0.029	0.215	0.88
x8	0.053	0.359	0.069	0.042	0.097	0.796	-0.009	0.01	0.782	0.777
x7	0.118	0.382	0.047	0.121	0.128	0.636	0.143	0.078	0.624	0.821
x6	0.07	0.164	0.17	0.042	0.126	0.122	0.769	0.043	0.686	0.787
x5	0.139	0.143	0.071	-0.009	0.202	-0.012	0.634	0.043	0.49	0.774
x17	0.101	0.082	0.151	0.075	0.075	0.048	0.058	0.748	0.617	0.769
x18	0.069	-0.005	0.234	0.075	-0.015	0.012	0.014	0.5	0.316	0.795

Table B7.2

8-Factor, 29 Variables, Quartimax Rotation

QUARTIMAX										
vars	PA1	PA2	PA3	PA5	PA4	PA7	PA6	PA8	communality	KMO MSAi
x35	0.897	0.072	0.059	0.057	0.065	-0.015	-0.022	0.021	0.823	0.813
x33	0.779	0.094	-0.05	0.015	0.071	0.039	0.075	-0.026	0.632	0.823
x32	0.764	0.116	0.185	0.118	0.147	-0.011	0.129	0.009	0.683	0.897
x36	0.64	0.099	0.079	0.113	0.015	0.047	0.01	0.087	0.448	0.867
x37	0.629	0.071	-0.002	0.043	-0.089	0.185	0.031	0.146	0.467	0.838
x4	0.092	0.876	0.025	0.051	0.08	0.151	0.061	0.038	0.813	0.828
x3	0.112	0.804	0.029	0.026	0.026	0.097	0.07	-0.028	0.676	0.838
x2	0.182	0.703	-0.004	-0.004	0.075	0.117	0.053	0.076	0.556	0.914
x11	0.078	0.484	0.156	0.067	0.003	0.266	0.154	0.008	0.363	0.901
x22	0.244	0.09	0.562	0.021	0.105	-0.019	0.143	0.087	0.423	0.832
x23	0.182	-0.038	0.515	0.055	0.081	0.083	0.041	0.012	0.318	0.828
x26	0.133	0.115	0.462	0.197	0.183	0.067	-0.021	0.15	0.344	0.809
x25	0.242	0.051	0.456	0.094	0.181	0.064	0.08	0.056	0.325	0.921
x24	0.232	0.083	0.426	0.05	0.08	0.061	0.098	0.171	0.294	0.898
x27	0.203	0.155	0.42	0.322	0.122	0.065	-0.023	0.181	0.398	0.842
x20	0.253	0.112	0.373	0.036	0.152	-0.069	0.124	0.205	0.303	0.853
x29	0.133	0.054	-0.018	0.864	0.034	0.052	0.034	0.067	0.777	0.691
x28	0.182	0.04	0.16	0.652	0.038	0.046	0.007	-0.003	0.489	0.748
x30	0.077	0.108	0.235	0.499	0	0.016	-0.01	0.031	0.324	0.789
x12	0.141	0.152	0.115	0.015	0.715	0.005	0.131	0.01	0.584	0.812
x13	0.302	0.121	0.096	0.083	0.677	0.073	0.194	0.006	0.624	0.857
x16	0.059	0.133	0.166	0.12	0.349	0.177	-0.003	0.162	0.242	0.862
x21	0.17	0.149	0.18	-0.009	0.32	0.157	0.049	0.043	0.215	0.88
x8	0.056	0.371	0.046	0.039	0.034	0.798	0.007	0.008	0.782	0.777
x7	0.128	0.4	0.005	0.119	0.068	0.634	0.148	0.071	0.624	0.821
x6	0.125	0.192	0.104	0.048	0.135	0.104	0.768	0.042	0.686	0.787
x5	0.179	0.168	-0.001	-0.004	0.206	-0.021	0.62	0.045	0.49	0.774
x17	0.138	0.101	0.107	0.096	0.038	0.05	0.062	0.748	0.617	0.769
x18	0.11	0.006	0.209	0.086	-0.03	0.013	0.026	0.501	0.316	0.795

Table B7.3

8-Factor, 29 Variables, Oblimin Rotation

OBLIMIN										
vars	PA1	PA2	PA3	PA5	PA4	PA7	PA6	PA8	communality	KMO MSAi
x35	0.902	0.021	0.03	0.005	0.03	-0.044	-0.065	-0.004	0.823	0.813
x33	0.798	0.014	-0.083	-0.02	0.037	0.019	0.046	-0.045	0.632	0.823
x32	0.691	0.038	0.153	0.061	0.082	-0.054	0.089	-0.032	0.683	0.897
x37	0.644	-0.057	-0.036	-0.01	-0.132	0.186	0.02	0.134	0.467	0.838
x36	0.616	0.033	0.048	0.065	-0.027	0.019	-0.019	0.065	0.448	0.867
x4	-0.033	0.881	-0.013	0.015	0.026	0.036	-0.007	0.018	0.813	0.828
x3	0.016	0.834	0.013	0	-0.036	-0.012	0.019	-0.052	0.676	0.838
x2	0.091	0.698	-0.046	-0.043	0.032	0.022	-0.007	0.065	0.556	0.914
x11	-0.031	0.379	0.148	0.026	-0.073	0.227	0.133	-0.028	0.363	0.901
x22	0.077	0.046	0.58	-0.059	0.01	-0.055	0.119	0.036	0.423	0.832
x23	0.054	-0.117	0.547	-0.012	0.016	0.093	0.022	-0.039	0.318	0.828
x25	0.088	-0.035	0.455	0.025	0.123	0.051	0.04	0.01	0.325	0.921
x26	-0.047	0.053	0.444	0.125	0.142	0.039	-0.079	0.11	0.344	0.809
x24	0.091	0.007	0.418	-0.027	0.009	0.037	0.07	0.137	0.294	0.898
x27	0.023	0.096	0.39	0.256	0.07	0.023	-0.076	0.138	0.398	0.842
x20	0.103	0.085	0.345	-0.035	0.09	-0.123	0.086	0.181	0.303	0.853
x29	0.007	-0.013	-0.097	0.89	0.009	0.004	0.022	0.027	0.777	0.691
x28	0.06	-0.015	0.126	0.654	-0.001	0.011	-0.008	-0.053	0.489	0.748
x30	-0.048	0.093	0.22	0.49	-0.047	-0.026	-0.024	-0.013	0.324	0.789
x12	-0.049	0.047	0.017	-0.027	0.764	-0.024	0.007	0	0.584	0.812
x13	0.119	-0.04	-0.009	0.037	0.706	0.053	0.082	-0.013	0.624	0.857
x16	-0.089	0.005	0.095	0.066	0.37	0.173	-0.085	0.152	0.242	0.862
x21	0.053	0.03	0.139	-0.062	0.321	0.157	-0.021	0.025	0.215	0.88
x8	-0.007	0.022	0.024	-0.022	0.017	0.881	-0.033	-0.023	0.782	0.777
x7	0.035	0.093	-0.046	0.066	0.028	0.669	0.114	0.043	0.624	0.821
x6	-0.039	-0.004	0.053	0.024	-0.012	0.053	0.815	0.006	0.686	0.787
x5	0.054	0.046	-0.064	-0.018	0.107	-0.081	0.641	0.027	0.49	0.774
x17	0	0.015	-0.025	0.001	0.017	-0.007	0.018	0.785	0.617	0.769
x18	0.005	-0.043	0.143	0.013	-0.064	-0.022	0.006	0.514	0.316	0.795

Even if an interpretable set of factors have already been achieved, additional runs were made for exploration.

Eight Run

The number of factors were reduced by one to compress the variables into larger groups of factors.

Table B8.1

7-Factor, 29 Variables, Varimax Rotation

VARIMAX									
vars	PA5	PA1	PA2	PA3	PA4	PA7	PA6	communality	KMO MSAi
x35	0.868	0.197	0.071	0.076	0.138	-0.039	0	0.823	0.813
x33	0.769	0.062	0.093	0.034	0.141	0.021	0.099	0.635	0.823
x32	0.703	0.288	0.102	0.129	0.226	-0.036	0.135	0.676	0.897
x37	0.62	0.176	0.073	0.049	-0.063	0.19	0.051	0.466	0.838
x36	0.61	0.218	0.094	0.122	0.053	0.04	0.031	0.449	0.867
x22	0.142	0.556	0.07	0.019	0.172	-0.041	0.095	0.375	0.832
x26	0.038	0.507	0.092	0.188	0.207	0.048	-0.047	0.35	0.809
x27	0.112	0.502	0.133	0.316	0.141	0.054	-0.039	0.406	0.842
x24	0.149	0.491	0.068	0.044	0.115	0.046	0.07	0.29	0.898
x20	0.166	0.478	0.093	0.03	0.154	-0.076	0.121	0.309	0.853
x25	0.152	0.459	0.031	0.089	0.24	0.036	0.044	0.303	0.921
x23	0.099	0.455	-0.048	0.052	0.163	0.046	-0.016	0.251	0.828
x18	0.059	0.439	-0.001	0.075	-0.115	0.045	0.054	0.22	0.795
x17	0.092	0.421	0.089	0.091	-0.062	0.092	0.105	0.225	0.769
x4	0.068	0.077	0.868	0.071	0.124	0.154	0.084	0.815	0.828
x3	0.095	0.046	0.798	0.047	0.088	0.096	0.081	0.674	0.838
x2	0.163	0.081	0.694	0.014	0.103	0.123	0.083	0.548	0.914
x11	0.046	0.166	0.471	0.072	0.069	0.262	0.134	0.348	0.901
x29	0.108	0.058	0.029	0.866	0.025	0.068	0.062	0.776	0.691
x28	0.139	0.177	0.019	0.65	0.074	0.041	0.002	0.481	0.748
x30	0.029	0.242	0.089	0.496	0.028	0.014	-0.018	0.314	0.789
x12	0.065	0.139	0.114	0.02	0.695	-0.022	0.159	0.546	0.812
x13	0.224	0.146	0.081	0.088	0.69	0.044	0.222	0.613	0.857
x21	0.115	0.208	0.129	-0.008	0.35	0.137	0.046	0.216	0.88
x16	0.001	0.241	0.109	0.116	0.328	0.167	0.012	0.219	0.862
x8	0.052	0.065	0.365	0.039	0.109	0.77	-0.016	0.747	0.777
x7	0.115	0.083	0.383	0.118	0.113	0.654	0.145	0.642	0.821
x6	0.071	0.178	0.167	0.033	0.156	0.118	0.714	0.614	0.787
x5	0.134	0.086	0.141	-0.009	0.196	-0.011	0.668	0.53	0.774

Table B8.2

7-Factor, 29 Variables, Quartimax Rotation

QUARTIMAX									
vars	PA1	PA2	PA3	PA5	PA4	PA7	PA6	communality	KMO MSAi
x35	0.898	0.073	0.06	0.057	0.061	-0.017	-0.023	0.823	0.813
x33	0.781	0.096	-0.064	0.018	0.074	0.037	0.071	0.635	0.823
x32	0.762	0.117	0.167	0.117	0.16	-0.015	0.118	0.676	0.897
x36	0.641	0.098	0.115	0.111	-0.008	0.051	0.019	0.449	0.867
x37	0.63	0.072	0.067	0.042	-0.13	0.196	0.044	0.466	0.838
x4	0.093	0.88	0.043	0.05	0.066	0.142	0.06	0.815	0.828
x3	0.112	0.806	0.01	0.026	0.036	0.084	0.056	0.674	0.838
x2	0.184	0.702	0.037	-0.004	0.047	0.116	0.063	0.548	0.914
x11	0.077	0.486	0.134	0.067	0.025	0.255	0.132	0.348	0.901
x22	0.244	0.089	0.522	0.022	0.142	-0.025	0.119	0.375	0.832
x26	0.134	0.116	0.496	0.189	0.179	0.064	-0.027	0.35	0.809
x27	0.204	0.155	0.475	0.315	0.106	0.065	-0.025	0.406	0.842
x24	0.234	0.085	0.456	0.047	0.082	0.059	0.091	0.29	0.898
x20	0.256	0.11	0.439	0.031	0.126	-0.063	0.137	0.309	0.853
x23	0.181	-0.031	0.437	0.057	0.139	0.064	0.007	0.251	0.828
x25	0.242	0.053	0.428	0.091	0.209	0.056	0.06	0.303	0.921
x18	0.118	0.005	0.415	0.083	-0.134	0.044	0.081	0.22	0.795
x17	0.153	0.099	0.387	0.098	-0.089	0.091	0.127	0.225	0.769
x29	0.134	0.056	0.021	0.866	0.017	0.057	0.038	0.776	0.691
x28	0.182	0.042	0.143	0.649	0.058	0.039	-0.011	0.481	0.748
x30	0.077	0.107	0.226	0.495	0.015	0.011	-0.02	0.314	0.789
x12	0.147	0.155	0.129	0.014	0.68	0.012	0.146	0.546	0.812
x13	0.306	0.125	0.105	0.082	0.662	0.078	0.204	0.613	0.857
x21	0.173	0.153	0.187	-0.011	0.319	0.157	0.047	0.216	0.88
x16	0.064	0.136	0.236	0.116	0.305	0.183	0.018	0.219	0.862
x8	0.055	0.385	0.04	0.04	0.044	0.768	-0.01	0.747	0.777
x7	0.128	0.407	0.037	0.119	0.051	0.649	0.144	0.642	0.821
x6	0.127	0.201	0.114	0.048	0.144	0.11	0.713	0.614	0.787
x5	0.179	0.17	0.02	0	0.189	-0.015	0.658	0.53	0.774

Table B8.3

7-Factor, 29 Variables, Oblimin Rotation

OBLIMIN									
vars	PA1	PA2	PA3	PA5	PA7	PA4	PA6	communality	KMO MSAi
x35	0.905	0.023	0.022	0.005	-0.048	0.034	-0.066	0.823	0.813
x33	0.8	0.017	-0.117	-0.014	0.017	0.048	0.042	0.635	0.823
x32	0.686	0.038	0.121	0.059	-0.061	0.11	0.075	0.676	0.897
x37	0.645	-0.061	0.054	-0.009	0.202	-0.179	0.042	0.466	0.838
x36	0.617	0.03	0.091	0.063	0.025	-0.049	-0.006	0.449	0.867
x4	-0.033	0.886	0.004	0.014	0.034	0.012	-0.003	0.815	0.828
x3	0.012	0.837	-0.025	-0.002	-0.023	-0.015	0.003	0.674	0.838
x2	0.093	0.697	0.005	-0.041	0.03	-0.002	0.012	0.548	0.914
x11	-0.037	0.375	0.112	0.021	0.218	-0.034	0.105	0.348	0.901
x22	0.072	0.037	0.547	-0.067	-0.066	0.071	0.086	0.375	0.832
x26	-0.049	0.049	0.505	0.109	0.037	0.14	-0.082	0.35	0.809
x24	0.088	0.003	0.477	-0.036	0.035	0.017	0.065	0.29	0.898
x27	0.021	0.092	0.476	0.241	0.023	0.055	-0.073	0.406	0.842
x18	0.019	-0.046	0.462	0.019	0.024	-0.21	0.095	0.22	0.795
x23	0.05	-0.114	0.457	-0.02	0.064	0.104	-0.028	0.251	0.828
x20	0.102	0.08	0.455	-0.044	-0.114	0.057	0.109	0.309	0.853
x25	0.084	-0.038	0.429	0.013	0.039	0.167	0.014	0.303	0.921
x17	0.037	0.02	0.417	0.031	0.065	-0.171	0.133	0.225	0.769
x29	0.007	-0.013	-0.068	0.895	0.009	-0.011	0.031	0.776	0.691
x28	0.056	-0.015	0.085	0.647	0	0.034	-0.034	0.481	0.748
x30	-0.052	0.087	0.2	0.479	-0.037	-0.019	-0.041	0.314	0.789
x8	-0.011	0.031	0	-0.025	0.858	0.034	-0.052	0.747	0.777
x7	0.032	0.079	-0.018	0.064	0.7	0.01	0.116	0.642	0.821
x12	-0.033	0.055	0.043	-0.021	-0.003	0.703	0.045	0.546	0.812
x13	0.128	-0.033	0.003	0.044	0.071	0.672	0.11	0.613	0.857
x21	0.057	0.031	0.153	-0.064	0.163	0.315	-0.015	0.216	0.88
x16	-0.078	0.007	0.204	0.066	0.192	0.302	-0.044	0.219	0.862
x6	-0.047	0.003	0.058	0.024	0.063	0.012	0.756	0.614	0.787
x5	0.042	0.04	-0.043	-0.007	-0.073	0.08	0.691	0.53	0.774

Ninth Run

The number of factors was reduced by one in the hopes of obtaining a more concise grouping of variables.

Table B9.1

6-Factor, 29 Variables, Varimax Rotation

VARIMAX								
vars	PA2	PA5	PA1	PA3	PA4	PA6	communality	KMO MSAi
x4	0.826	0.077	0.066	0.036	0.112	0.095	0.715	0.828
x3	0.734	0.103	0.037	0.013	0.081	0.095	0.566	0.838
x2	0.676	0.167	0.07	-0.013	0.092	0.09	0.506	0.914
x7	0.637	0.095	0.09	0.136	0.093	0.085	0.458	0.821
x8	0.637	0.033	0.073	0.075	0.078	-0.04	0.426	0.777
x11	0.56	0.041	0.16	0.065	0.054	0.118	0.362	0.901
x35	0.052	0.872	0.195	0.069	0.136	0.011	0.824	0.813
x33	0.103	0.77	0.061	0.032	0.136	0.102	0.637	0.823
x32	0.088	0.706	0.286	0.122	0.225	0.141	0.673	0.897
x36	0.112	0.61	0.216	0.12	0.051	0.029	0.449	0.867
x37	0.165	0.604	0.175	0.061	-0.062	0.027	0.432	0.838
x22	0.05	0.146	0.554	0.013	0.171	0.105	0.371	0.832
x26	0.115	0.038	0.506	0.184	0.204	-0.046	0.348	0.809
x27	0.155	0.114	0.5	0.31	0.137	-0.039	0.403	0.842
x24	0.092	0.149	0.492	0.043	0.11	0.068	0.291	0.898
x20	0.054	0.171	0.473	0.02	0.155	0.131	0.298	0.853
x25	0.055	0.151	0.46	0.089	0.239	0.04	0.304	0.921
x23	-0.011	0.097	0.457	0.058	0.162	-0.023	0.248	0.828
x18	0.025	0.057	0.44	0.078	-0.117	0.048	0.219	0.795
x17	0.131	0.089	0.42	0.093	-0.068	0.097	0.225	0.769
x29	0.075	0.107	0.058	0.866	0.024	0.056	0.774	0.691
x28	0.049	0.139	0.177	0.651	0.075	-0.002	0.482	0.748
x30	0.094	0.031	0.24	0.487	0.026	-0.012	0.305	0.789
x12	0.11	0.066	0.139	0.012	0.694	0.166	0.545	0.812
x13	0.118	0.221	0.148	0.087	0.685	0.221	0.61	0.857
x21	0.195	0.111	0.207	-0.005	0.343	0.034	0.212	0.88
x16	0.193	-0.004	0.241	0.121	0.32	-0.001	0.212	0.862
x5	0.134	0.132	0.084	-0.012	0.183	0.709	0.579	0.774
x6	0.229	0.068	0.178	0.037	0.152	0.665	0.556	0.787

Table B9.2

6-Factor, 29 Variables, Quatimax Rotation

QUARTIMAX								
vars	PA1	PA2	PA5	PA3	PA4	PA6	communality	KMO MSAi
x35	0.901	0.059	0.058	0.054	0.058	-0.014	0.824	0.813
x33	0.781	0.108	-0.064	0.018	0.07	0.073	0.637	0.823
x32	0.763	0.106	0.166	0.113	0.159	0.121	0.673	0.897
x36	0.64	0.118	0.114	0.111	-0.009	0.013	0.449	0.867
x37	0.618	0.163	0.073	0.052	-0.123	0.013	0.432	0.838
x4	0.096	0.833	0.028	0.033	0.065	0.072	0.715	0.828
x3	0.115	0.739	-0.003	0.009	0.037	0.074	0.566	0.838
x2	0.183	0.682	0.023	-0.018	0.045	0.07	0.506	0.914
x7	0.118	0.645	0.052	0.134	0.054	0.066	0.458	0.821
x8	0.049	0.641	0.052	0.071	0.04	-0.056	0.426	0.777
x11	0.072	0.569	0.13	0.066	0.021	0.109	0.362	0.901
x22	0.245	0.075	0.519	0.019	0.142	0.123	0.371	0.832
x26	0.134	0.138	0.493	0.188	0.179	-0.033	0.348	0.809
x27	0.204	0.174	0.471	0.312	0.106	-0.03	0.403	0.842
x24	0.233	0.111	0.457	0.048	0.081	0.083	0.291	0.898
x23	0.179	0.008	0.439	0.061	0.14	-0.008	0.248	0.828
x20	0.257	0.076	0.434	0.025	0.127	0.144	0.298	0.853
x25	0.241	0.08	0.429	0.092	0.211	0.05	0.304	0.921
x18	0.116	0.031	0.418	0.085	-0.133	0.07	0.219	0.795
x17	0.15	0.141	0.389	0.1	-0.09	0.113	0.225	0.769
x29	0.133	0.082	0.022	0.865	0.016	0.033	0.774	0.691
x28	0.181	0.059	0.142	0.65	0.059	-0.015	0.482	0.748
x30	0.078	0.103	0.222	0.489	0.014	-0.014	0.305	0.789
x12	0.148	0.151	0.126	0.011	0.68	0.15	0.545	0.812
x13	0.305	0.159	0.106	0.084	0.659	0.198	0.61	0.857
x21	0.169	0.218	0.186	-0.007	0.318	0.027	0.212	0.88
x16	0.061	0.215	0.236	0.121	0.303	-0.004	0.212	0.862
x5	0.177	0.161	0.021	0	0.18	0.699	0.579	0.774
x6	0.125	0.256	0.123	0.05	0.146	0.66	0.556	0.787

Table B9.2

6-Factor, 29 Variables, Quatimax Rotation

OBLIMIN								
vars	PA2	PA1	PA3	PA5	PA4	PA6	communality	KMO MSAi
x4	0.844	-0.018	-0.018	-0.013	0.021	0.019	0.715	0.828
x3	0.75	0.027	-0.045	-0.029	-0.004	0.03	0.566	0.838
x2	0.683	0.101	-0.014	-0.062	0.003	0.025	0.506	0.914
x8	0.671	-0.036	0.023	0.031	0.024	-0.108	0.426	0.777
x7	0.636	0.008	0.008	0.097	0.013	0.024	0.458	0.821
x11	0.551	-0.043	0.109	0.021	-0.034	0.081	0.362	0.901
x35	-0.029	0.914	0.017	-0.001	0.027	-0.055	0.824	0.813
x33	0.026	0.803	-0.118	-0.014	0.039	0.043	0.637	0.823
x32	-0.022	0.692	0.118	0.052	0.106	0.081	0.673	0.897
x37	0.116	0.63	0.062	0.006	-0.176	0.005	0.432	0.838
x36	0.047	0.62	0.089	0.061	-0.052	-0.01	0.449	0.867
x22	-0.031	0.078	0.545	-0.075	0.072	0.096	0.371	0.832
x26	0.066	-0.046	0.503	0.104	0.145	-0.083	0.348	0.809
x24	0.026	0.089	0.481	-0.036	0.016	0.06	0.291	0.898
x27	0.097	0.026	0.472	0.233	0.059	-0.072	0.403	0.842
x18	-0.024	0.02	0.469	0.021	-0.211	0.087	0.219	0.795
x23	-0.062	0.049	0.461	-0.015	0.107	-0.043	0.248	0.828
x20	-0.027	0.11	0.449	-0.054	0.06	0.122	0.298	0.853
x25	-0.014	0.084	0.431	0.013	0.171	0.005	0.304	0.921
x17	0.077	0.035	0.423	0.034	-0.173	0.123	0.225	0.769
x29	-0.005	0.008	-0.067	0.895	-0.011	0.034	0.774	0.691
x28	-0.02	0.058	0.083	0.647	0.037	-0.032	0.482	0.748
x30	0.045	-0.047	0.195	0.469	-0.017	-0.025	0.305	0.789
x12	0.036	-0.035	0.037	-0.025	0.709	0.047	0.545	0.812
x13	0.018	0.122	0.003	0.049	0.673	0.1	0.61	0.857
x21	0.163	0.05	0.153	-0.057	0.318	-0.039	0.212	0.88
x16	0.163	-0.086	0.205	0.074	0.304	-0.068	0.212	0.862
x5	-0.022	0.033	-0.037	0.001	0.058	0.741	0.579	0.774
x6	0.082	-0.051	0.076	0.034	0.015	0.698	0.556	0.787